

Reaction types in Organometallic Chemistry

—Elementary Reactions in Organometallic Chemistry (基元反应)

- (1) Ligand coordination and substitution ✓
(配体的配位和取代)
- (2) Oxidative addition and reductive elimination ✓
(氧化加成和还原消除)
- (3) Insertion and elimination (de-insertion) reactions
(插入反应和消除(反插入)反应)
- (4) Electrophilic and Nucleophilic Addition and Abstraction
(亲电和亲核加成与攫取)
(Reactions on the complexed ligands 和金属结合的配体的反应)

Help us to understand and predict the metal-catalyzed or mediated reactions.

Be aware:

- Classified in terms of reaction patterns, not mechanism !!
- An elementary reaction may go through different mechanisms.

Chapter 7. Insertion and Elimination

Outline

A. Introduction

B. CO insertion reactions

C. Insertion reactions involving alkene and alkyne

D. α , β , γ and δ elimination reactions

References and suggested readings:

1. *The Organometallic Chemistry of the Transition Metals*, Robert H. Crabtree, 6th Edition, **2014**, Chapter 6.
2. *Organometallic Chemistry*, G. O. Spessard, G. L. G. L. Miessler, Prentice-Hall: New Jersey, **1997**, Chapter 8.
3. *Advanced Inorganic Chemistry*, F. A. Cotton, G. Wilkinson, John Wiley & Sons, 5th ed. **1988**, Chapter 27, 1186.
4. *Organotransition Metal Chemistry*, Akio Yamamoto, **1986**, 246-272.

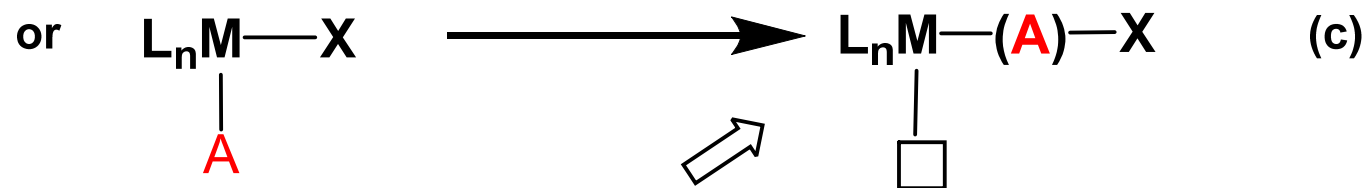
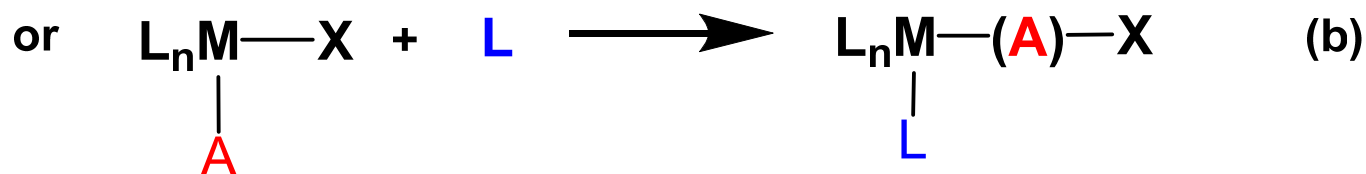
A. Introduction

➤ **Insertion reactions** can be represented by the following equation.



A: CO, carbene, alkene, alkyne, allene, CO₂, R₂C=O, O=O, etc

Insertion reactions often include the following processes:

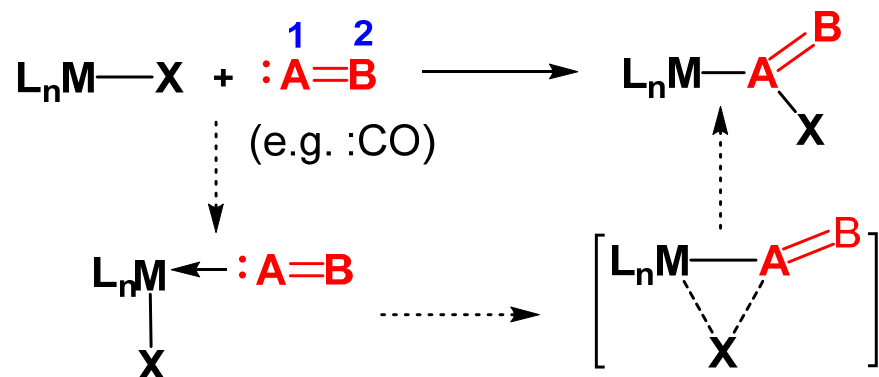


May be involved in (a) and (b)

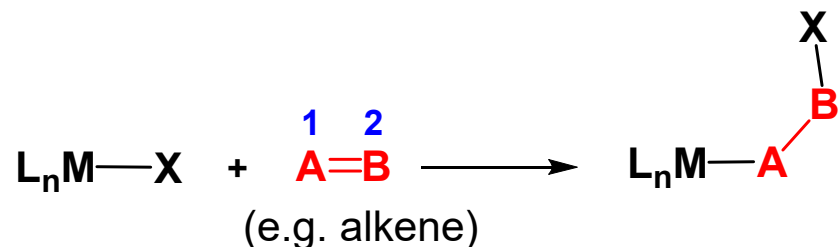
A. Introduction

➤ Common types of insertion reactions:

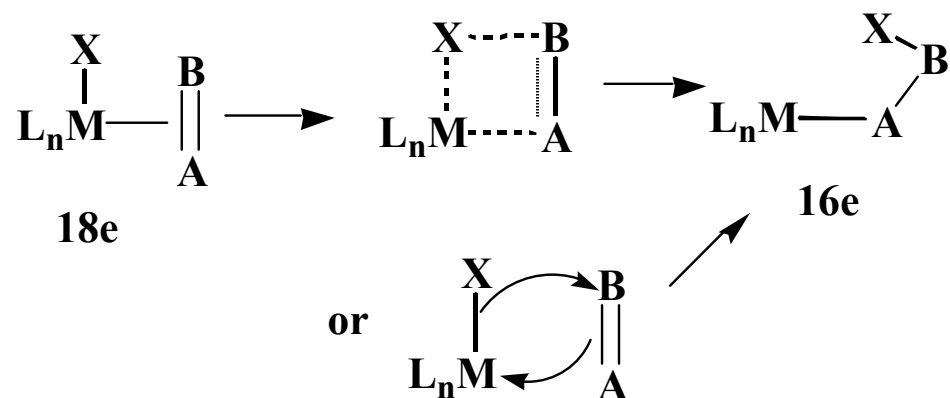
✓ 1,1-insertion:



✓ 1,2-insertion:

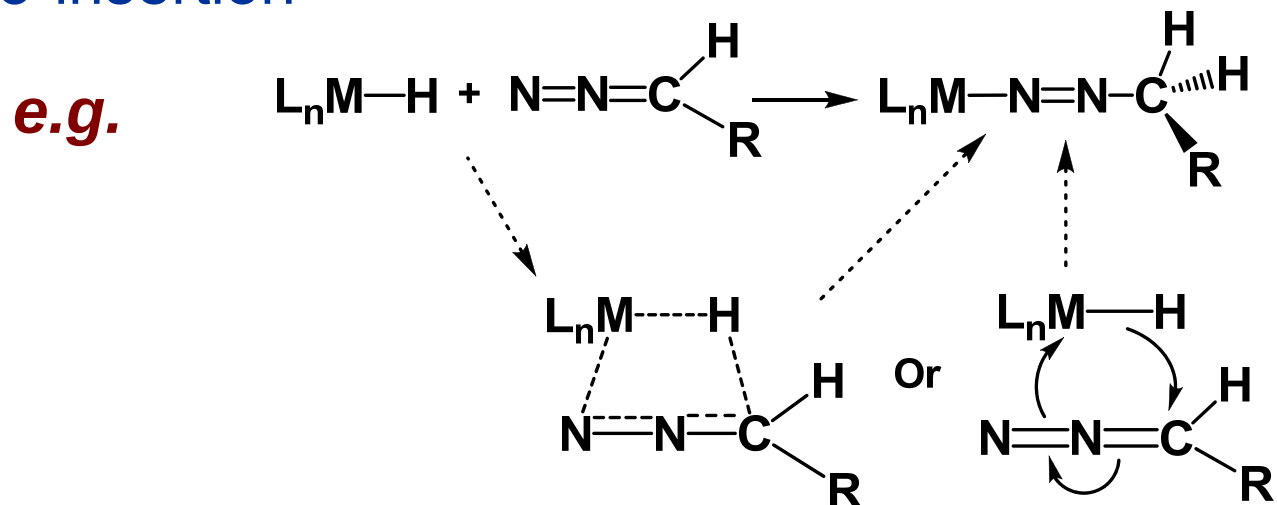


Often involve the following steps:

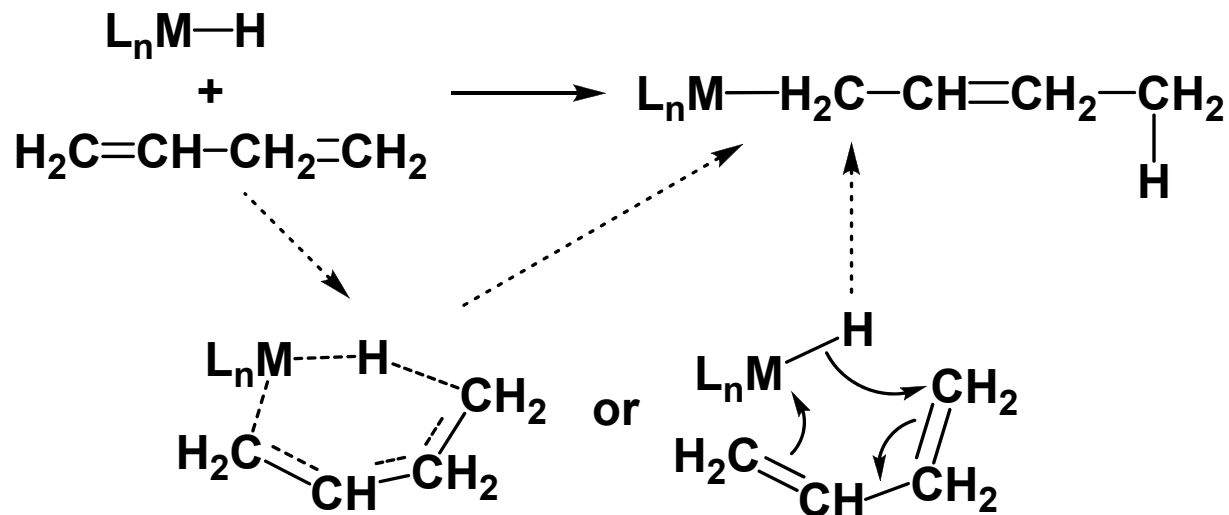


Other less common insertion reactions:

- 1,3-insertion



- 1,4-insertion

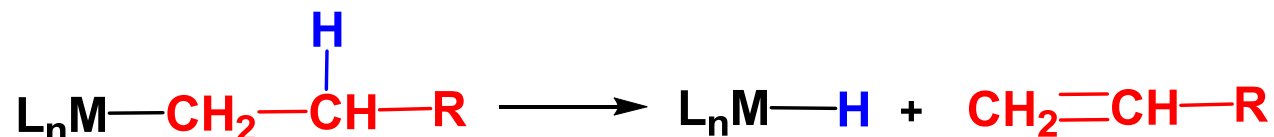
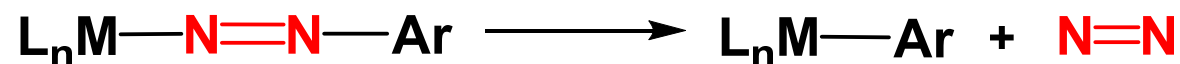


A. Introduction

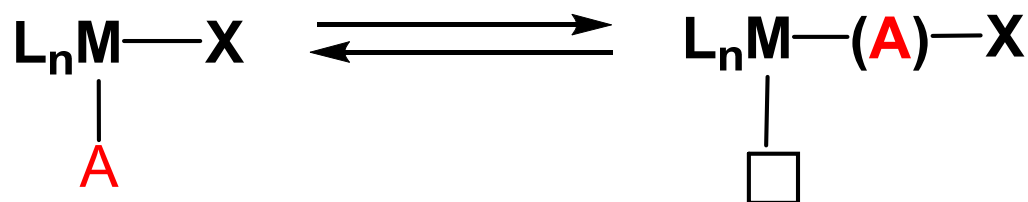
- **Elimination reactions** are just reverse reactions of insertion, can be also called **de-insertion** or **extrusion reactions**.



e.g.

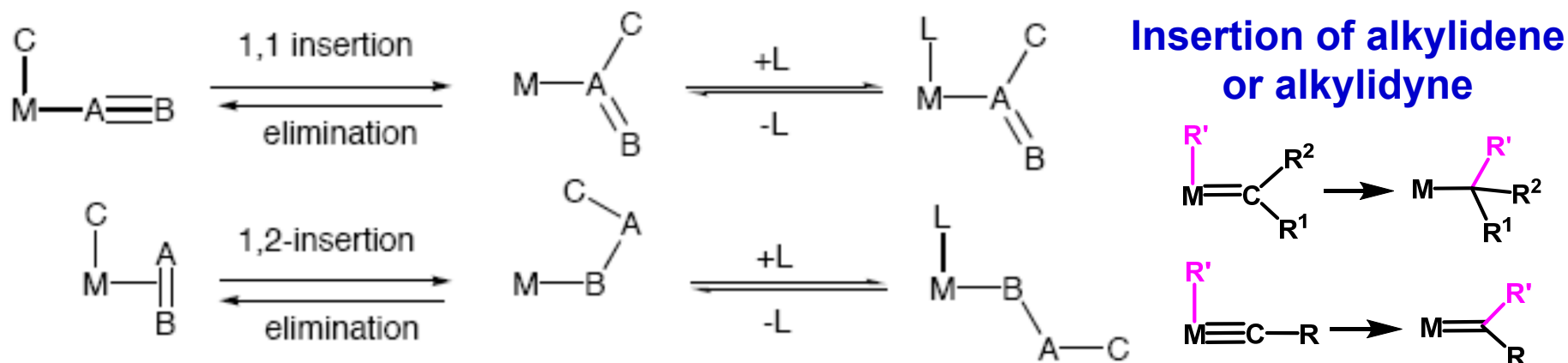


In principle:



- Main focus of this chapter: mechanisms of the insertion and elimination reactions (CO, alkene).

General features for insertion and elimination reactions

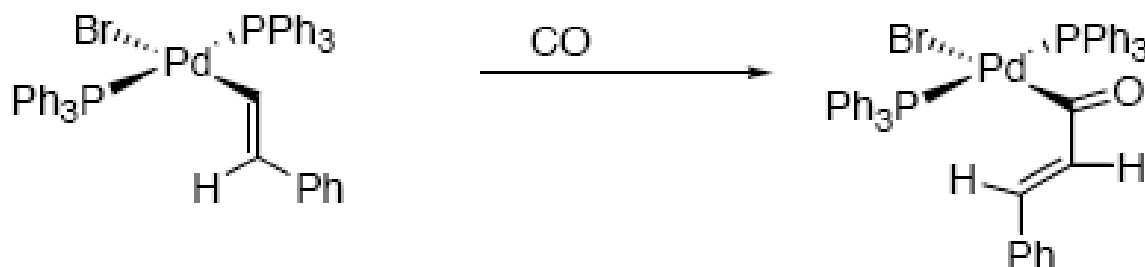
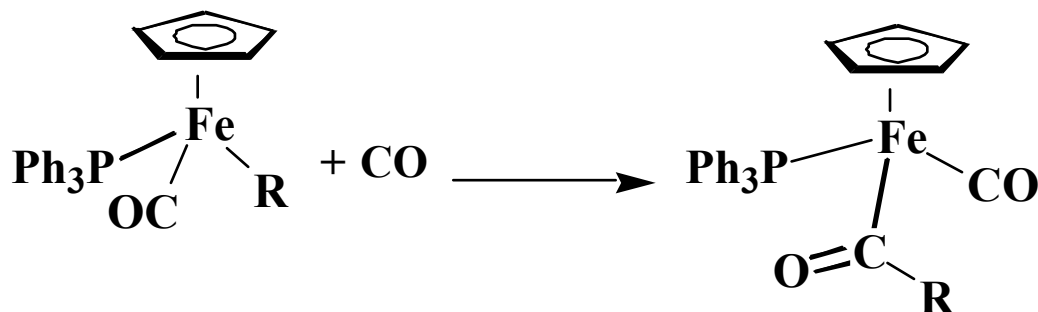
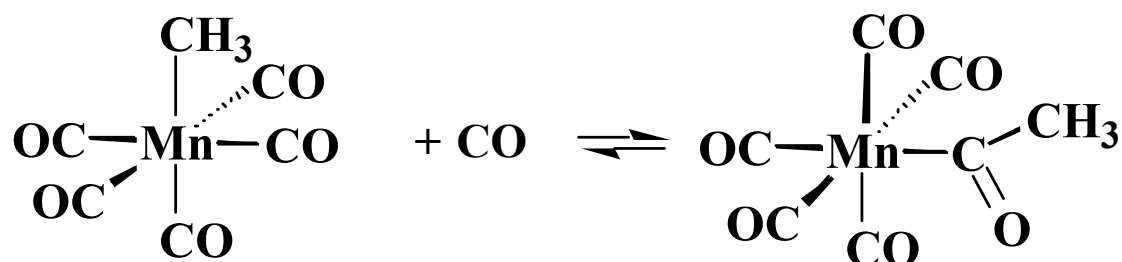


1. There is **no change** in the formal oxidation state of the metal unless AB is an alkylidene or alkylidyne.
2. The groups undergoing migratory insertion must be **cis** to one another. In complexes where the **cis** coordination sites are blocked by strongly coordinated ligands insertion or elimination processes are **not possible**.
3. An open coordination site is created during migratory insertion. Therefore, for the reverse reaction (elimination) to occur, an open coordination site must be generated by ligand dissociation.
4. In cases where C is a chiral carbon center, the reaction usually occurs with **retention of configuration**.
5. Cases where C migrates to AB followed by coordination of L in place of C, and where AB migrates to C followed by coordination of L in place of AB are both known.

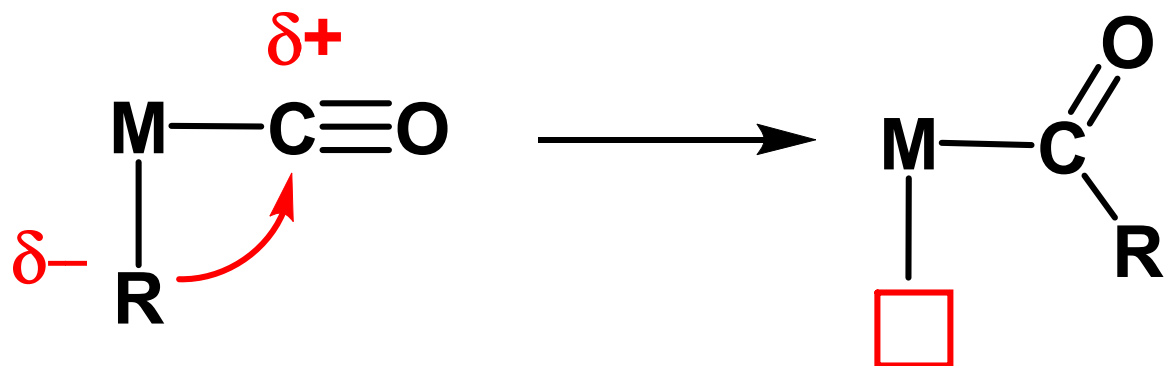
B. Insertion and de-insertion reactions involving CO and related substrates

- CO insertion is a typical example of 1,1-insertion reaction.

There are many examples of CO insertions. *e.g.*

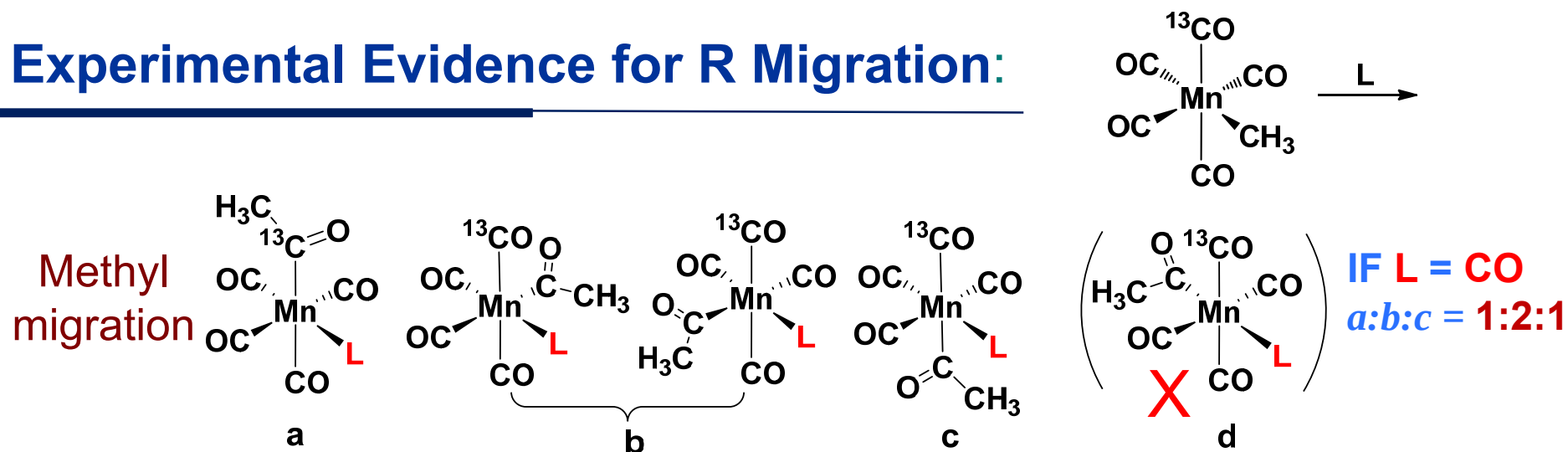


1) Main features of CO insertion reactions

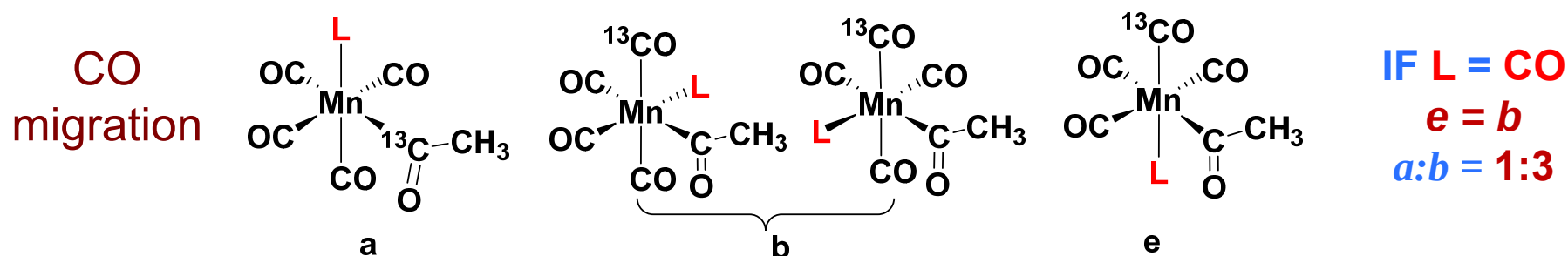


- a). CO and R **must be *cis***.
- b). It **can be described as R migration**
(or as an intramolecular nucleophilic addition reaction)
- c). The **configuration of R is unchanged**.
- d). The feasibility and rate depend on M-R bond strength.

Experimental Evidence for R Migration:



Methyl migration to CO followed by ligand coordination at the position vacated by Me cannot give **d**.

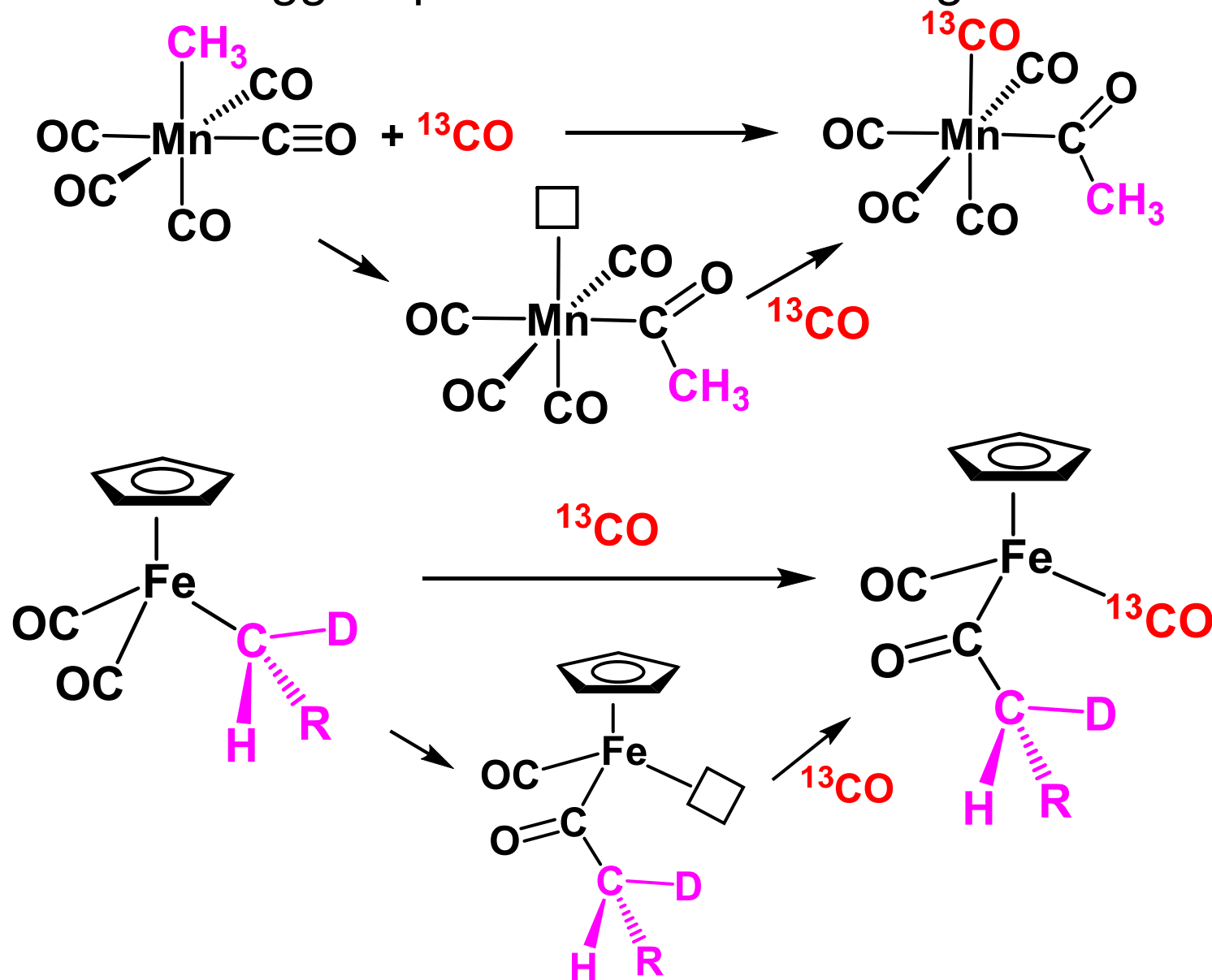


IF the incoming ligand is CO, a 1:2:1 ratio of **a:b:c** would be expected for *methyl migration*, while in the case of *CO migration*, **e** is also the isomer of **b**, thus a 1:3 ratio of **a:b** would be expected.

The experimental results (1:2:1) are consistent with methyl migration.

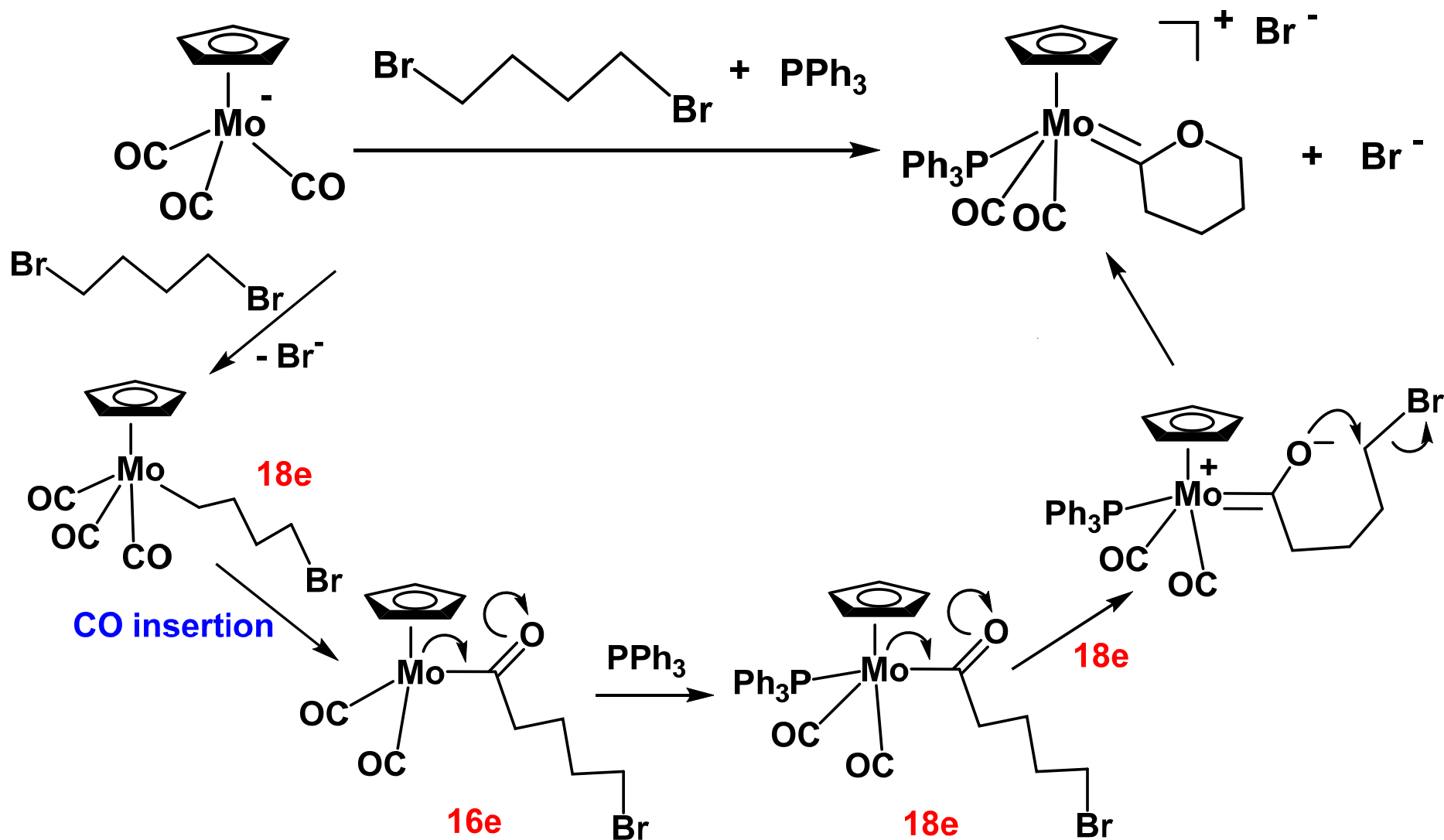
Practice Problems

Exercise 1. Suggest product for the following reactions

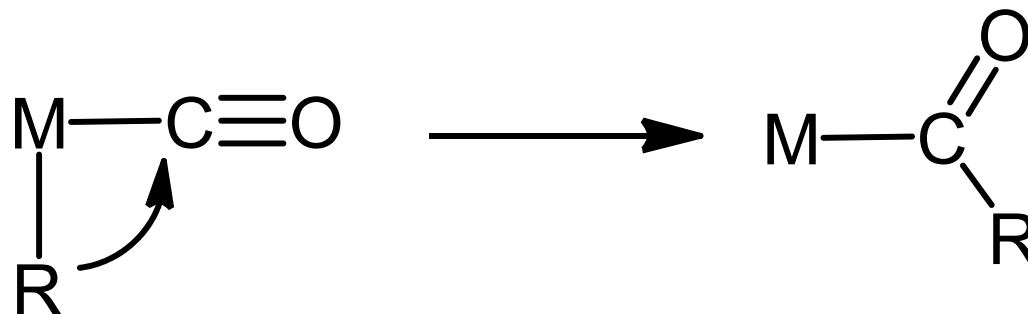


Practice Problems

Exercise 2. Propose a mechanism for the reaction:



2). Thermodynamics of CO insertions

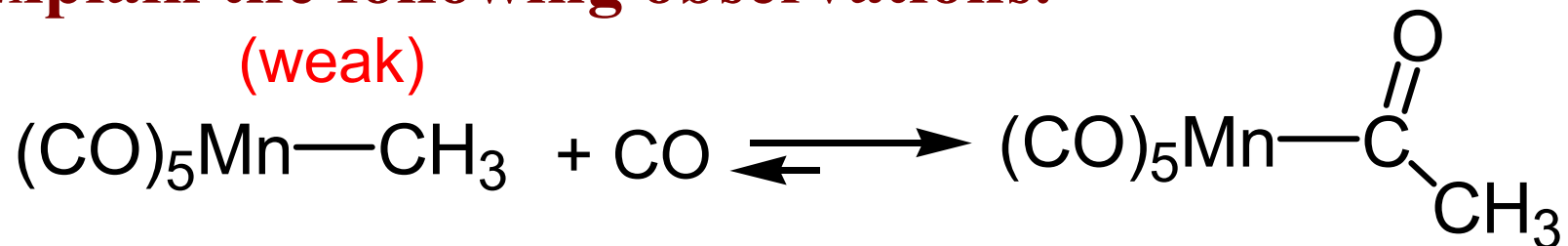


- Bond strength of M-R is a determining factor.
- The weaker the M-R bond, the easier to undergo a insertion reaction and the more favorable thermodynamically.

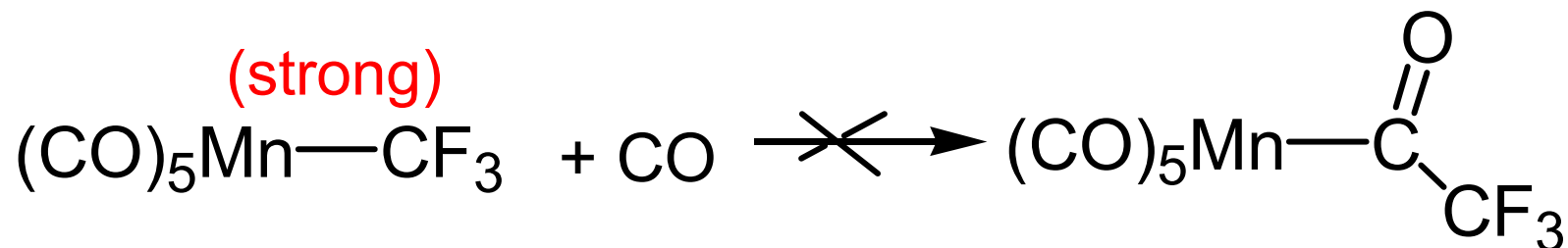
Practice Problems

Explain the following observations.

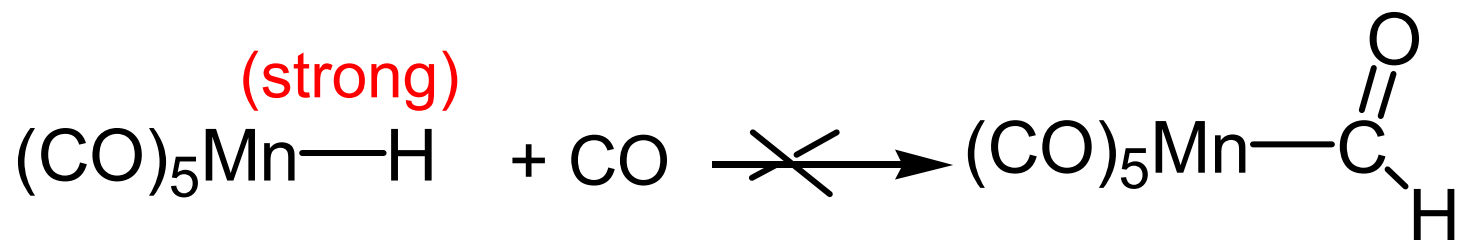
(weak)



(strong)

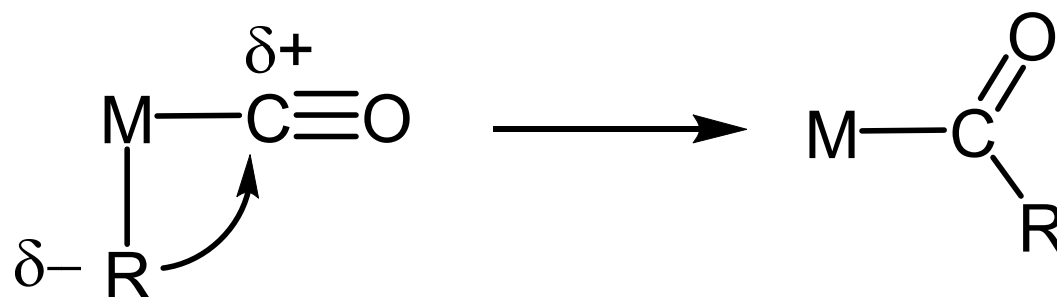


(strong)



- **Bond strength of M-R is a determining factor.**
- **Strong M-R bond disfavors insertion product.**
- **Relative bond strength: $\text{M-H} > \text{M-CF}_3 > \text{M-CH}_3$**

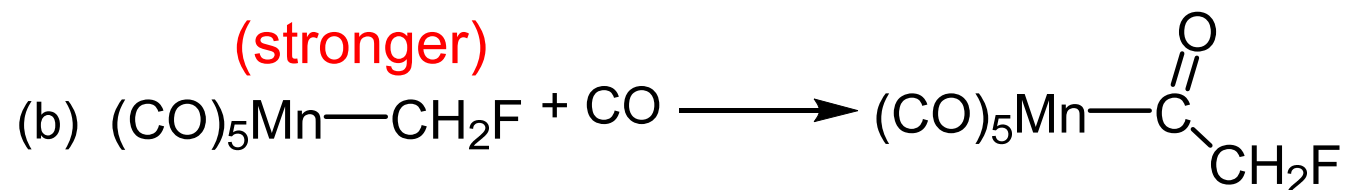
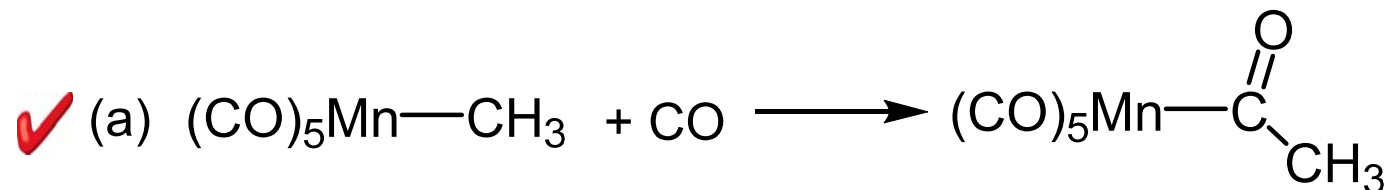
3). Factors influencing the insertion reaction rates



- ✓ The reaction can be viewed as an intramolecular nucleophilic addition reaction
- ✓ **General expectation: rate will increase when**
 - a) The positive charge on C atom of CO is increased;
 - b) M-R bond strength is weaker.

Practice Problems

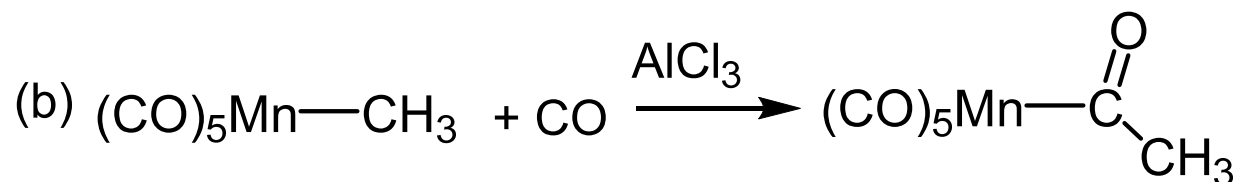
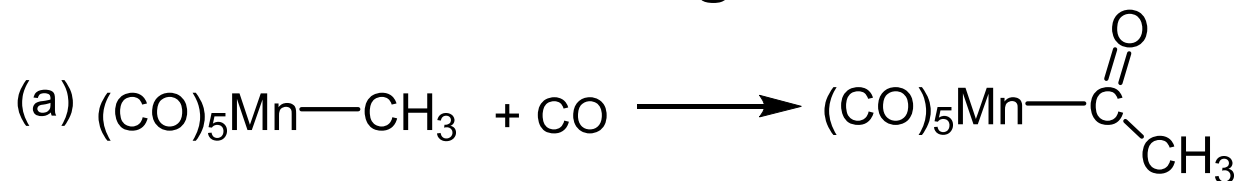
Q1. Which of the following reaction is faster?



Answer: (a), Because Mn-CH₃ bond is weaker.

Practice Problems

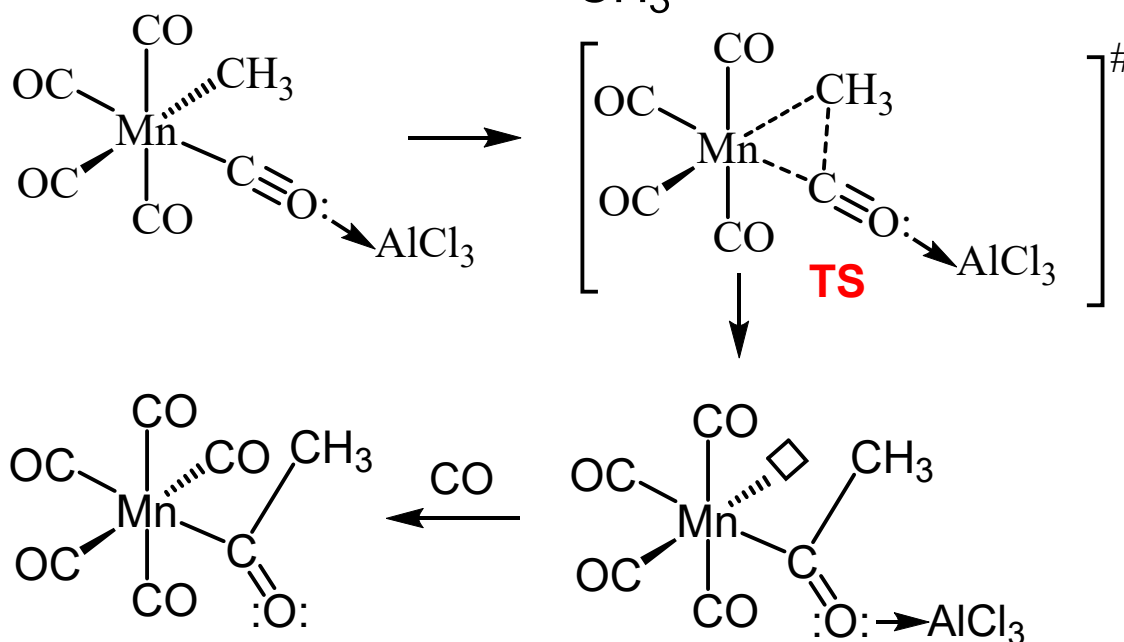
Q2. Which of the following reaction is faster?



Answer: (b)

Explanation:

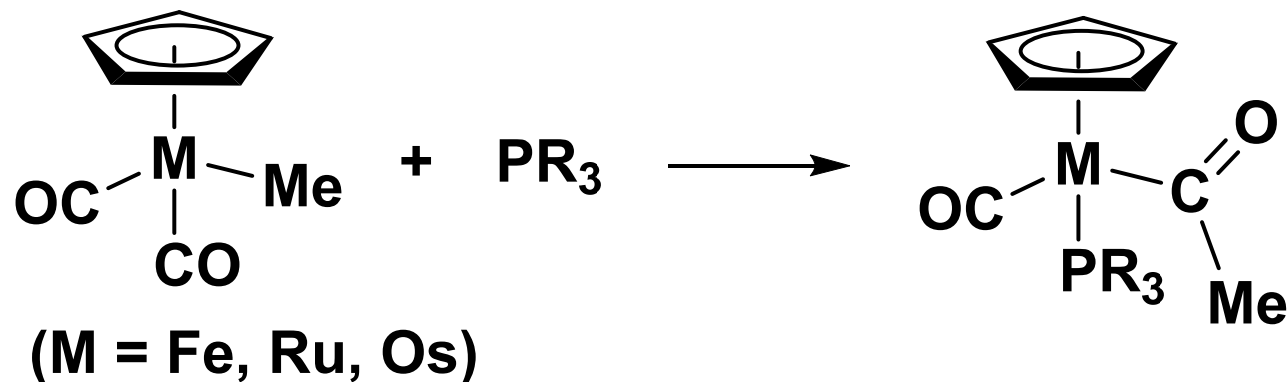
Lewis acid accelerate the reaction by reducing the charge on C of CO.



Similarly, cations such as Na^+ , Mg^{2+} can also enhance the insertion reaction rate.

Practice Problems

Q3. Rank the rate of the following reactions:

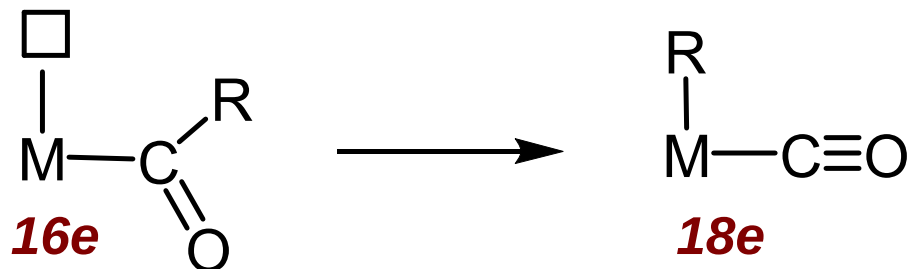


The transition metal-alkyl bond strength increases when going down a group.

M-Me bond strengths are in the order of $\text{Fe} < \text{Ru} < \text{Os}$

The order of the reaction rate: $\text{Fe} > \text{Ru} \gg \text{Os}$

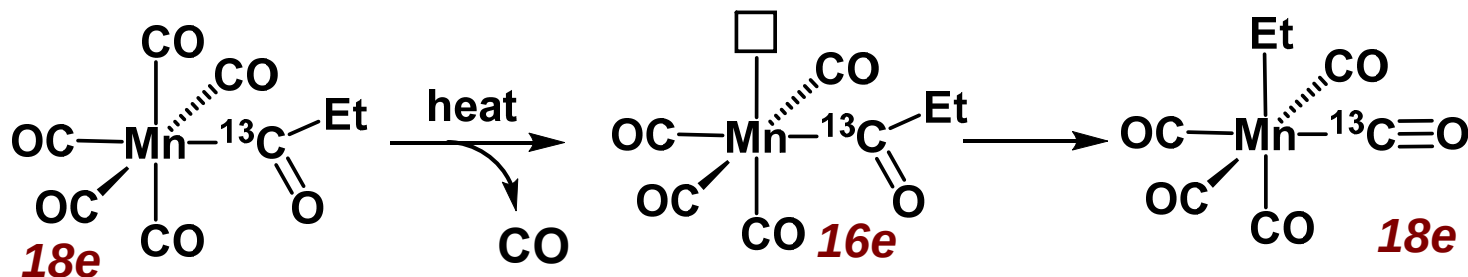
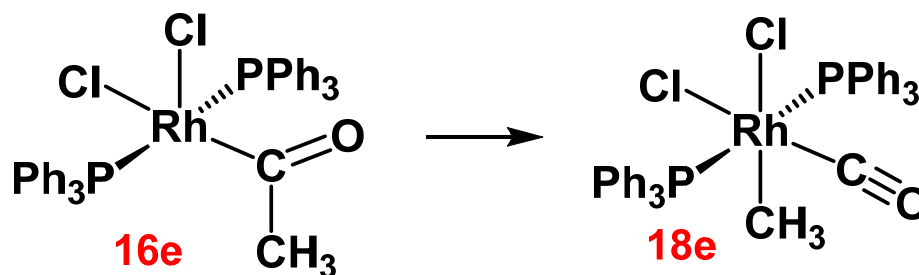
4). Decarbonylation reactions (CO de-insertion)



- The complexes must be unsaturated coordinatively.
- The CO and R are *cis* to each other in the products.
- The rate is dependent on R.

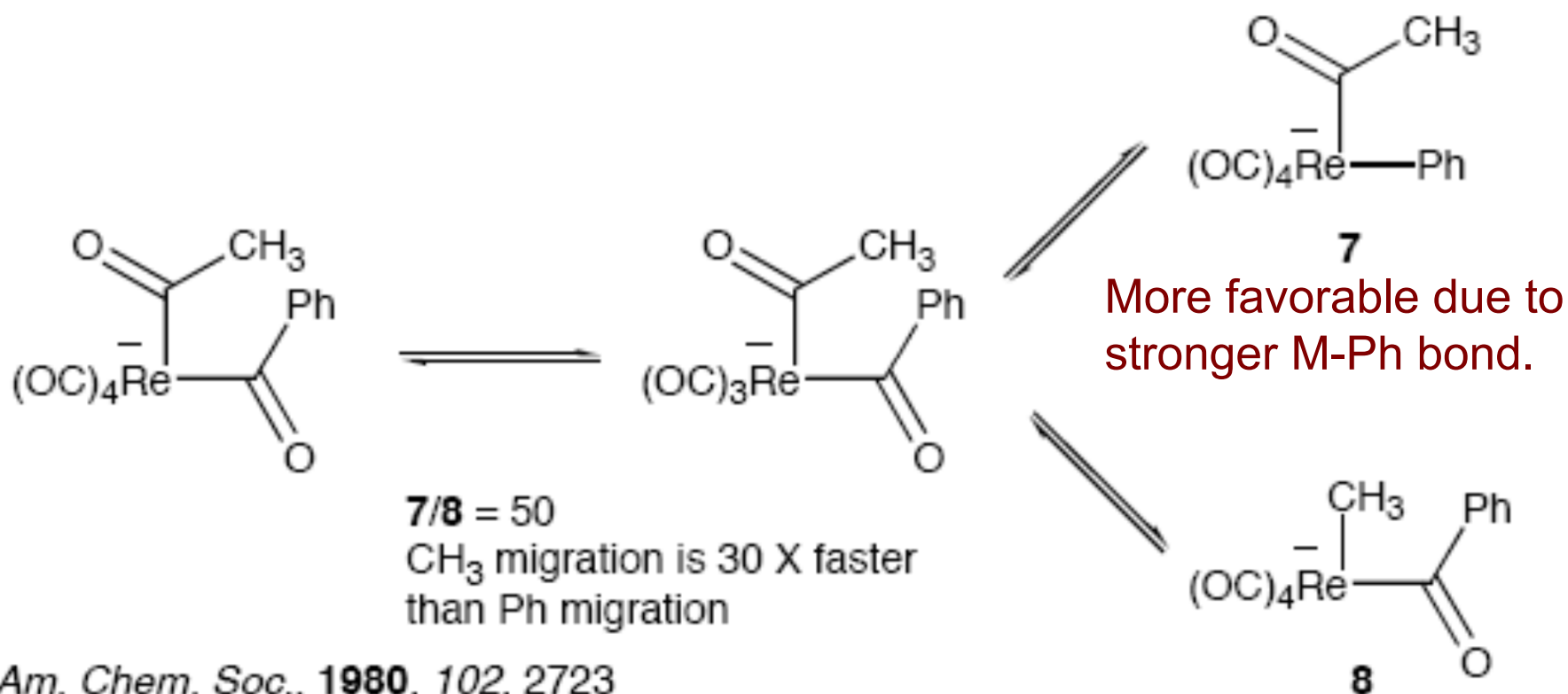
(The reaction is faster if the M-R bond is stronger)

Examples



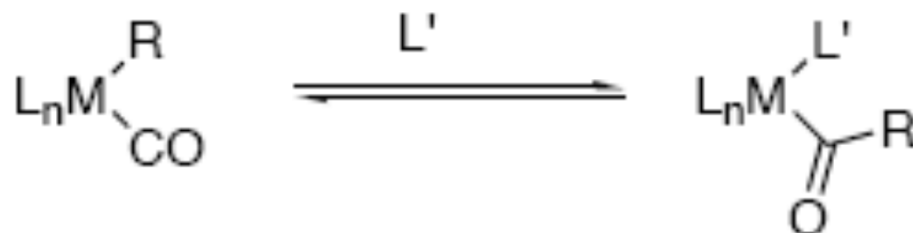
(One CO ligand leaves from metal center firstly)

An example of Effect of R on rate of decarbonylation reaction



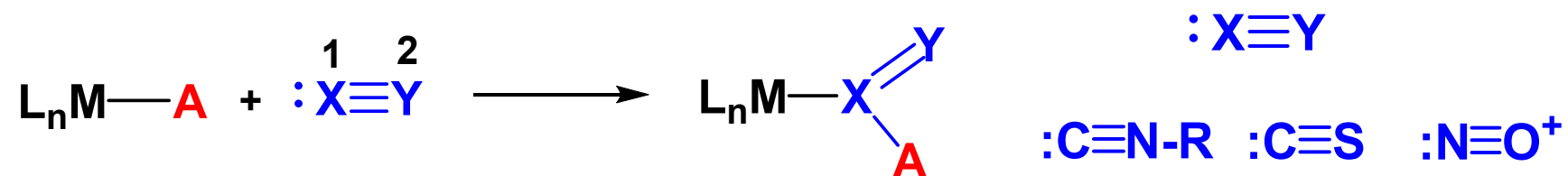
J. Am. Chem. Soc., **1980**, 102, 2723

Cautious: The identity of R has a significant effect on the aptitude to undergo migratory insertion to form an metal acyl. **However, Migratory insertion ability may not be correlated to M-R bond strength in some cases.**

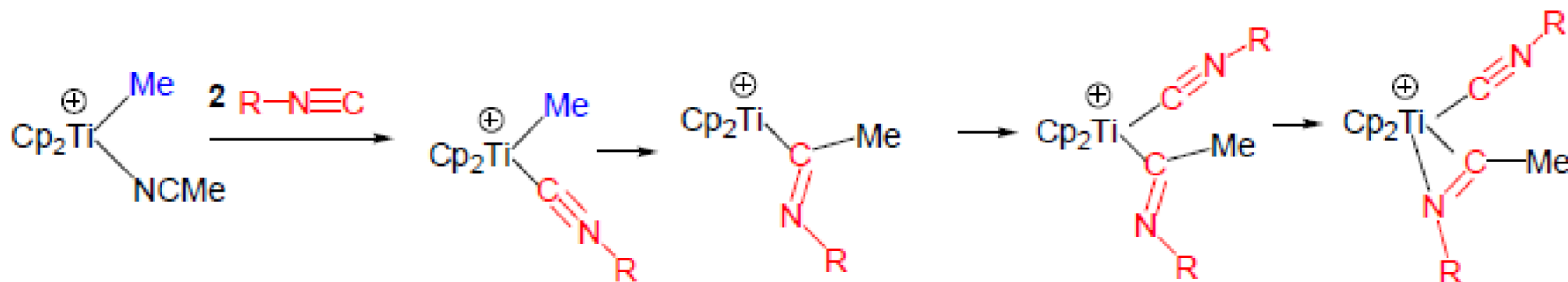
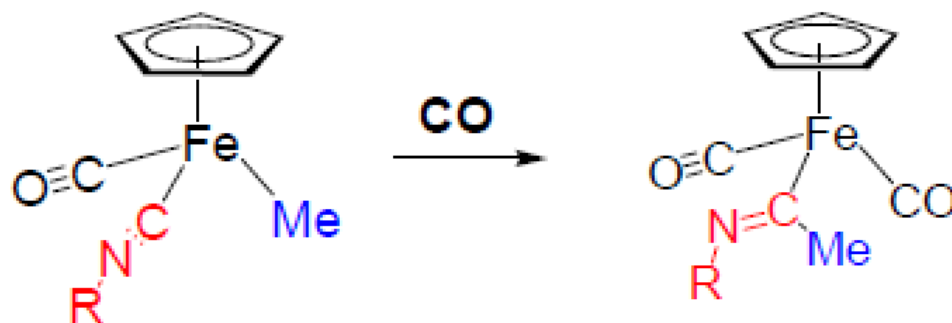


R	M-L BDE	Insertion/Elimination
PhCH ₂	29	Slower insertion than CH ₃
CH ₃ CO-	39	Elimination
CH ₃	44	Fast insertion
CF ₃	49	Elimination
Ph	49	Slower insertion than CH ₃ , but the product is more stable
H	59	Elimination

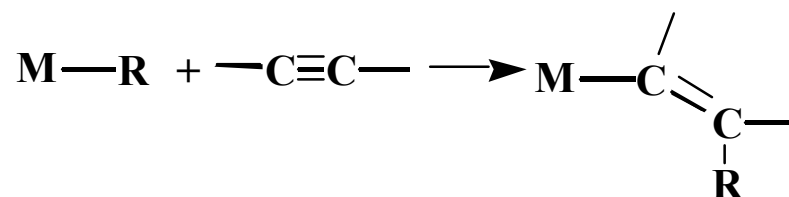
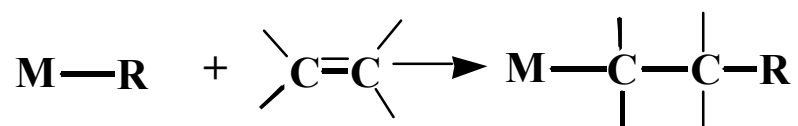
5). Related insertion reactions



Examples



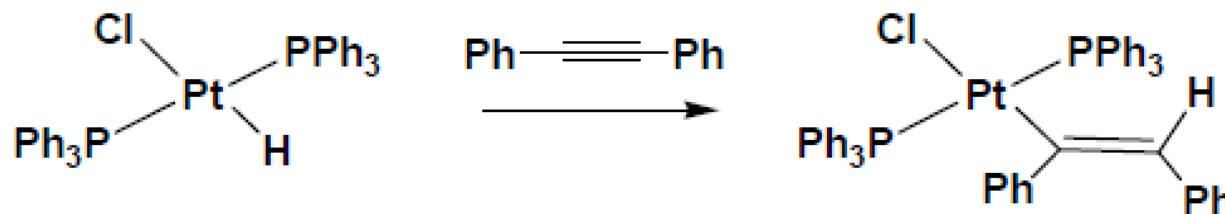
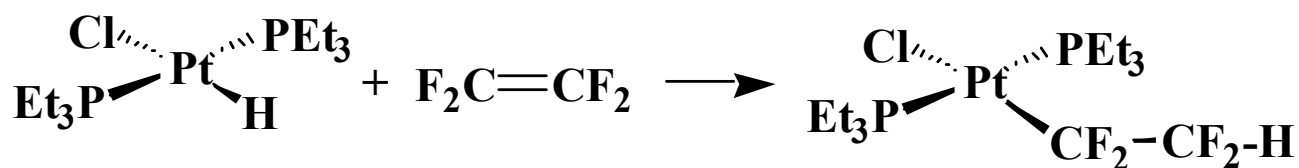
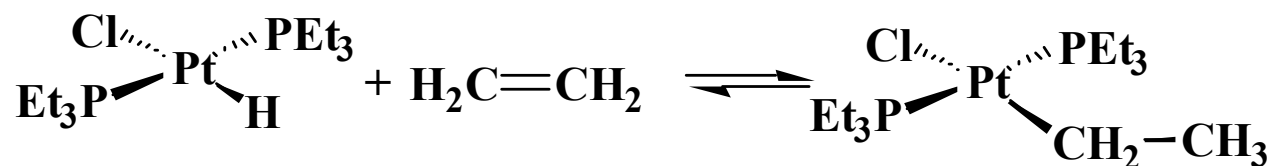
C. Insertion reactions involving alkenes and alkynes

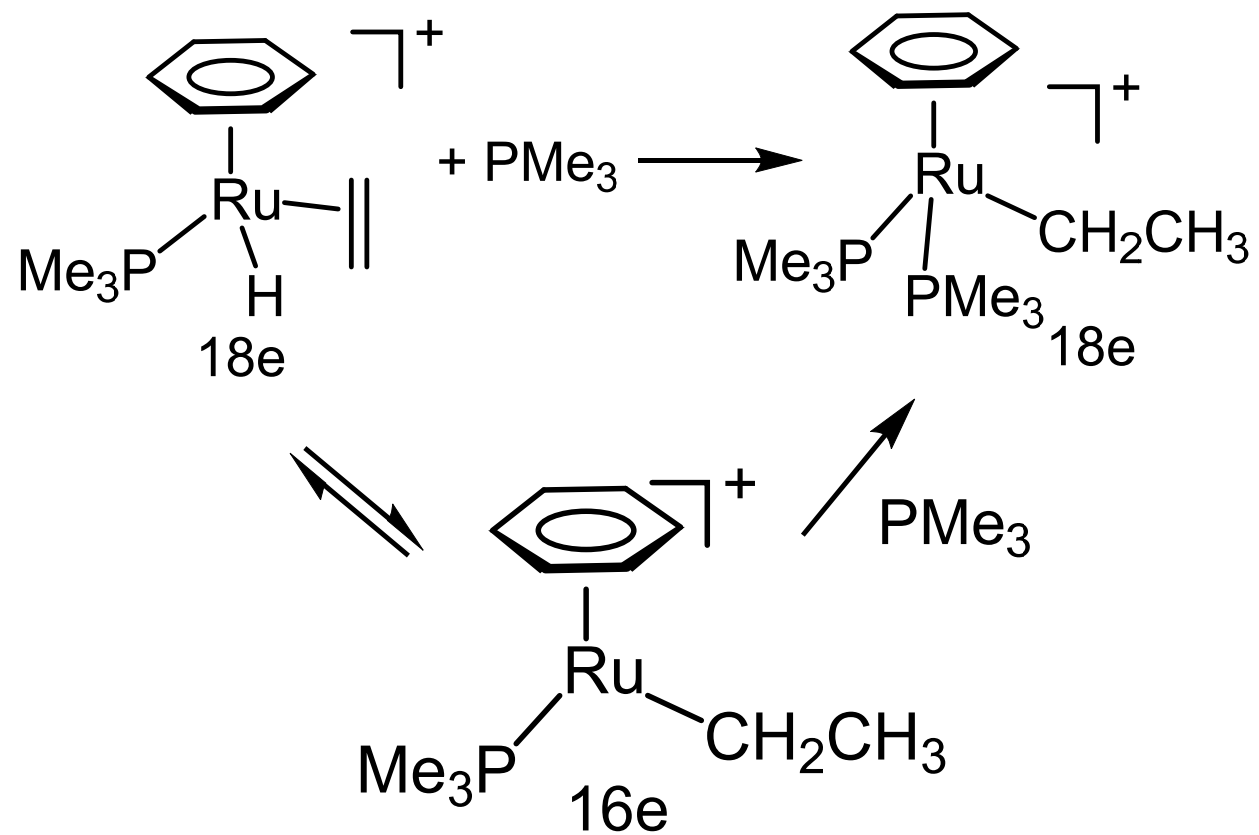


✓ **Very important in catalysis.**

✓ **Examples of 1,2-insertion reaction.**

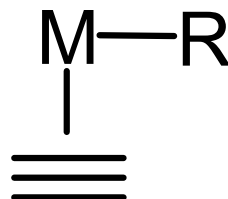
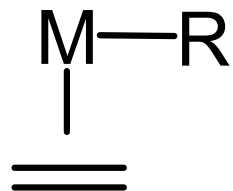
1) Examples of alkene and alkyne insertion reactions



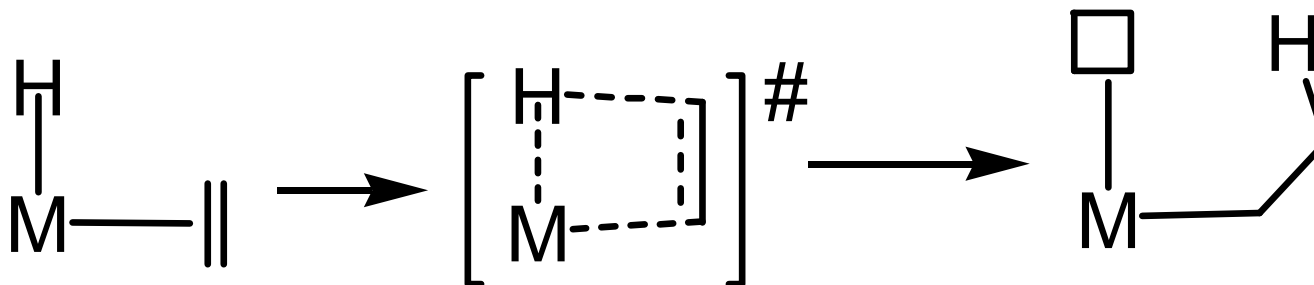


2). Some features of alkene and alkyne insertion reactions

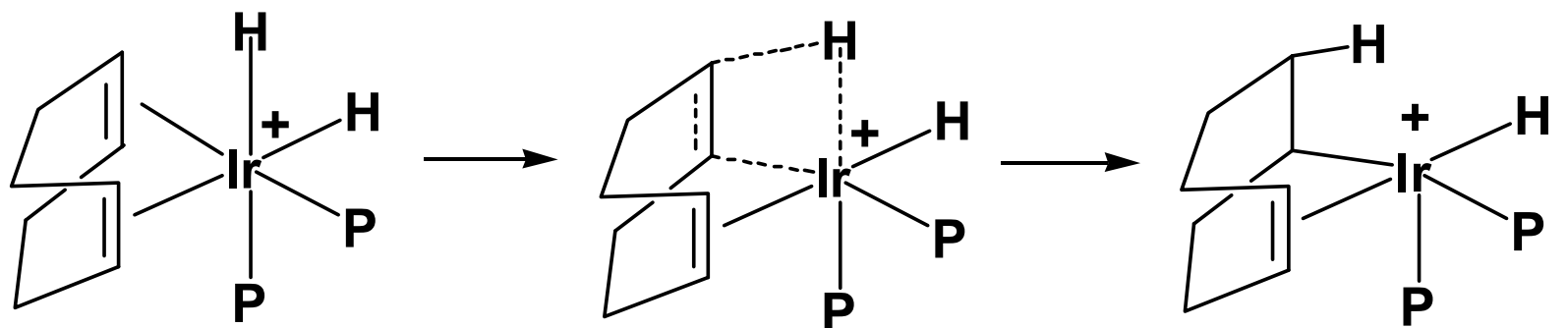
a). Need an intermediate in which the olefin or alkyne is *cis* to R.

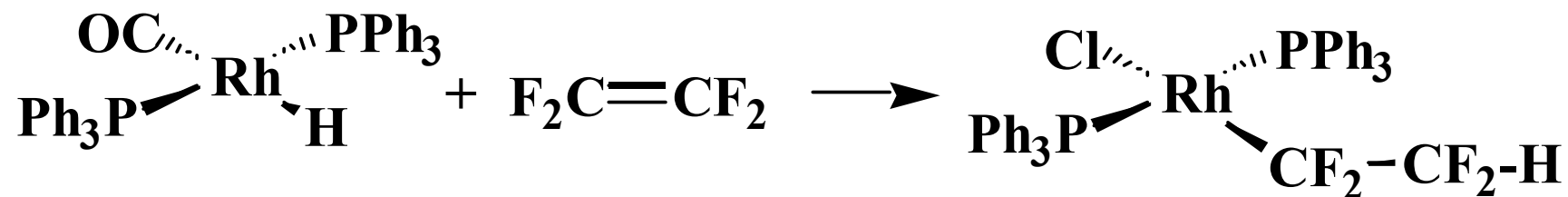


b). The transition state has a coplanar geometry.



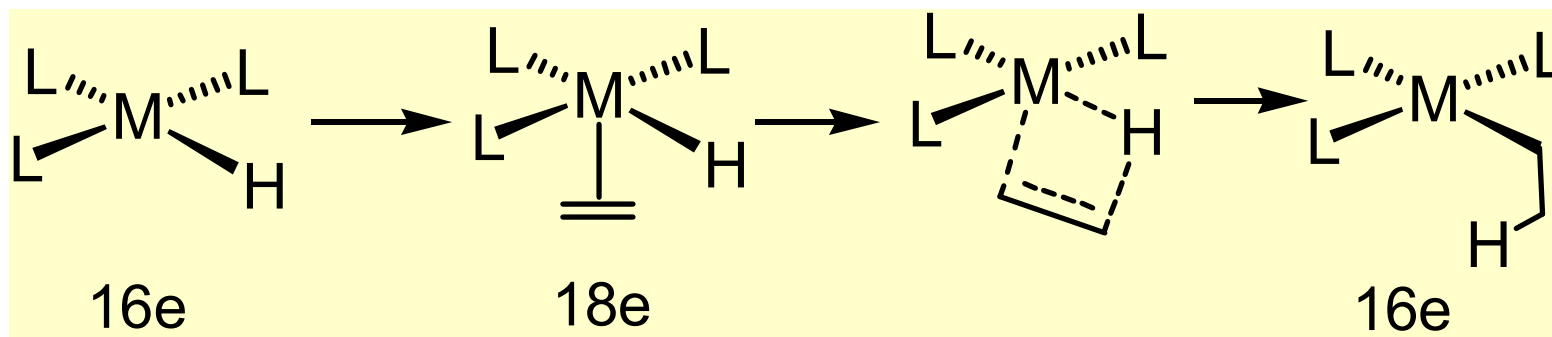
Example:



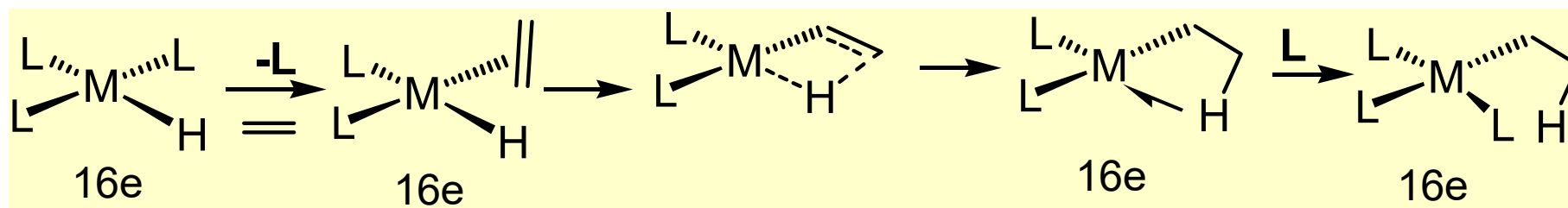


Mechanism?

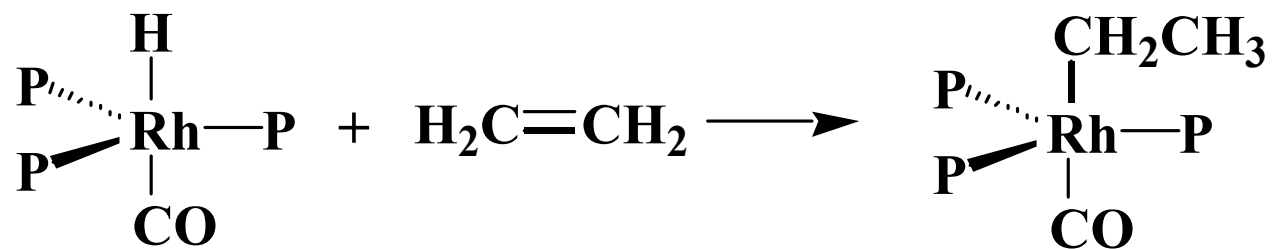
Possibility A:



Possibility B:

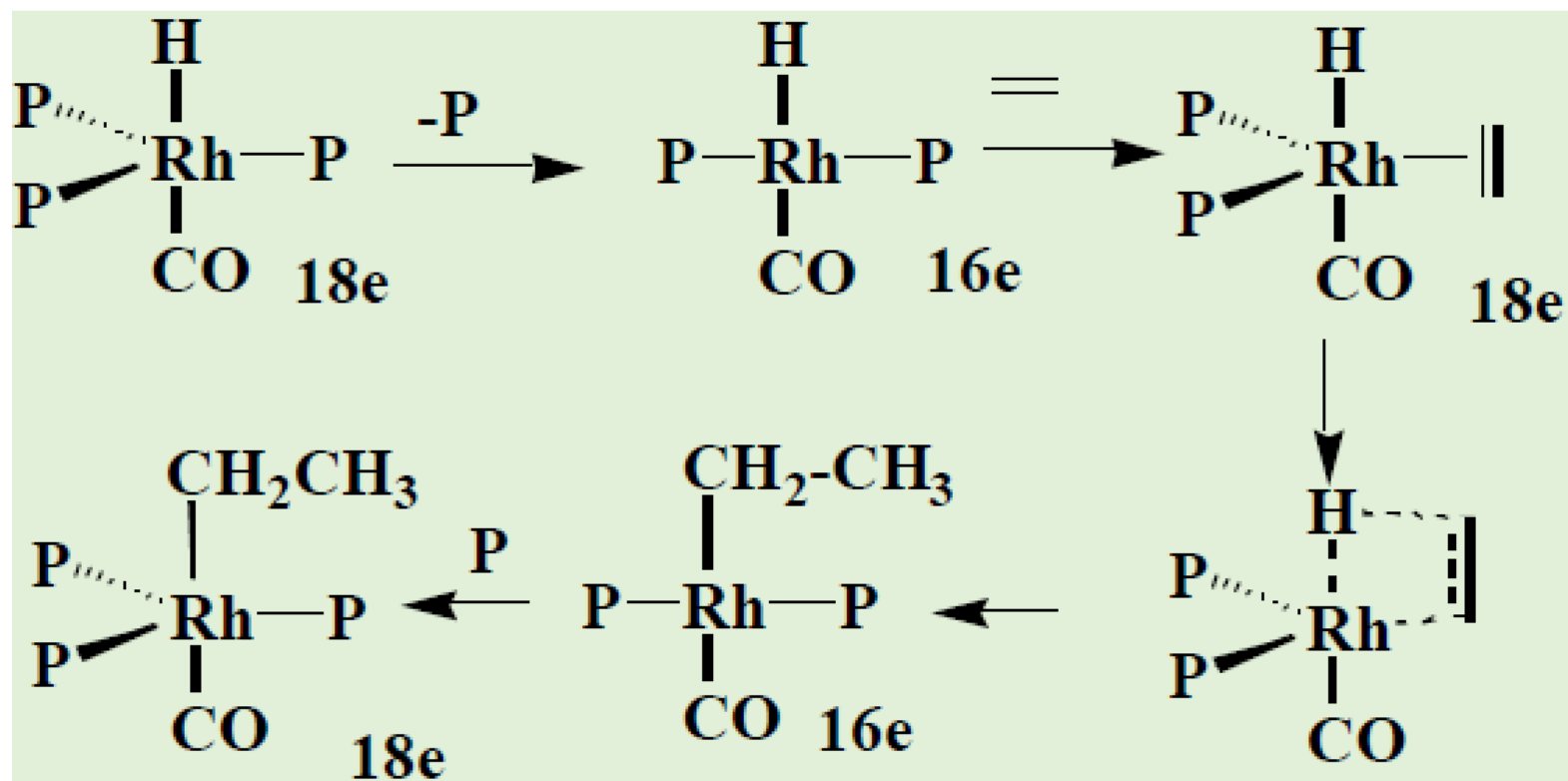


More consistent with experimental data.

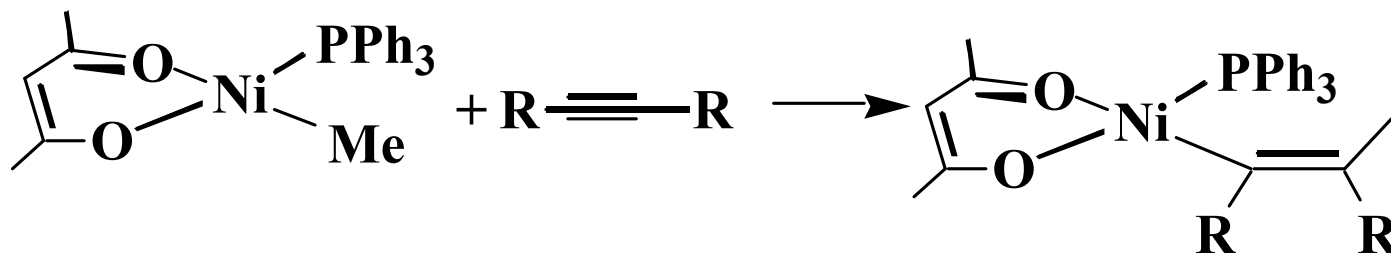
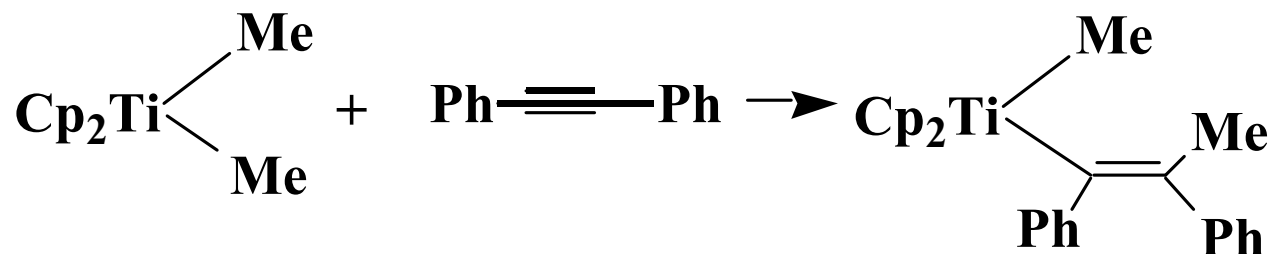
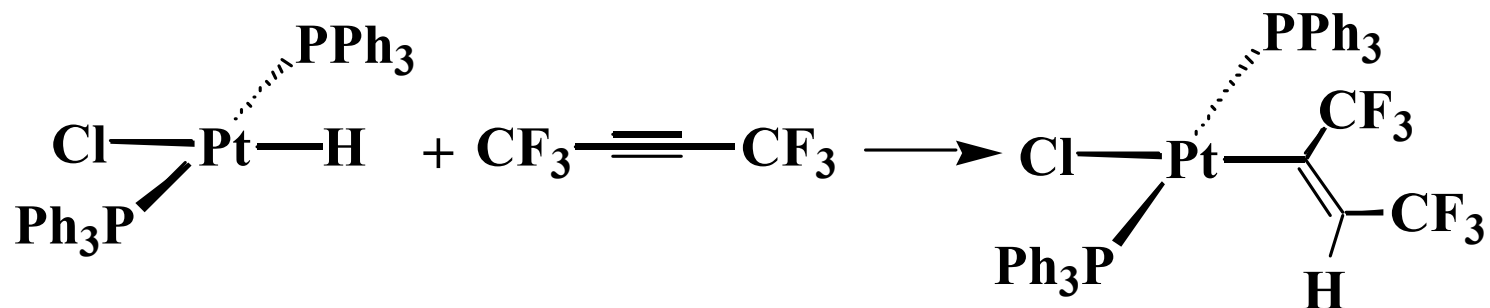


$\text{P} = \text{PPh}_3, 18\text{e}$

Mechanism?

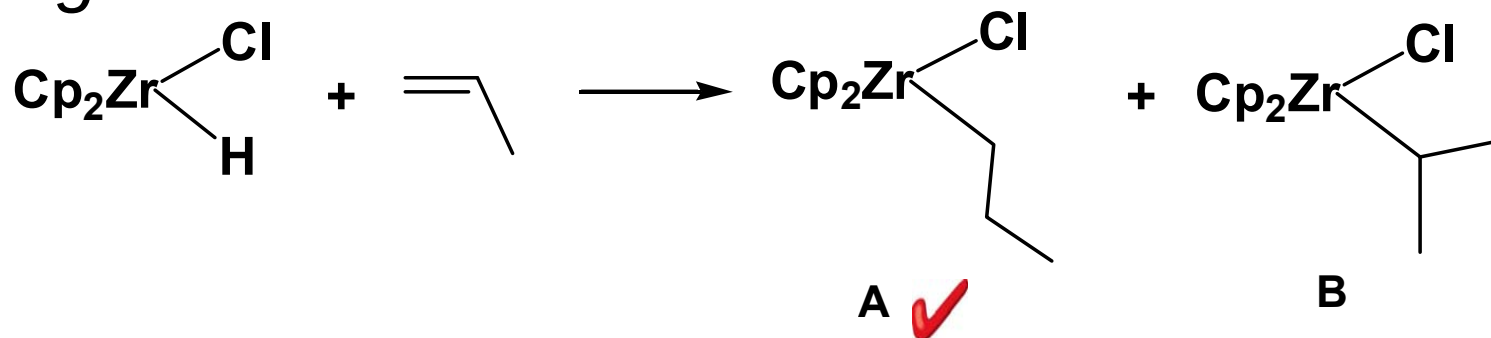


c). The stereochemistry is *syn* for the two substituents of $C\equiv C$. *e.g.*

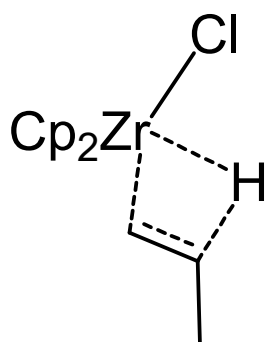


d). For unsymmetrically substituted substrates, the reactions can potentially give a mixture of isomers. The selectivity can be determined by both steric and electronic effect.

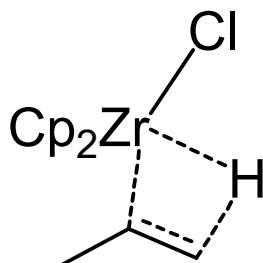
e.g.



Which one is major ?



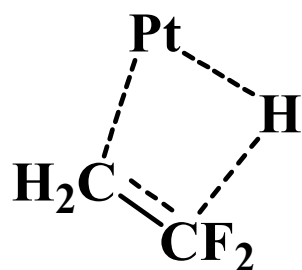
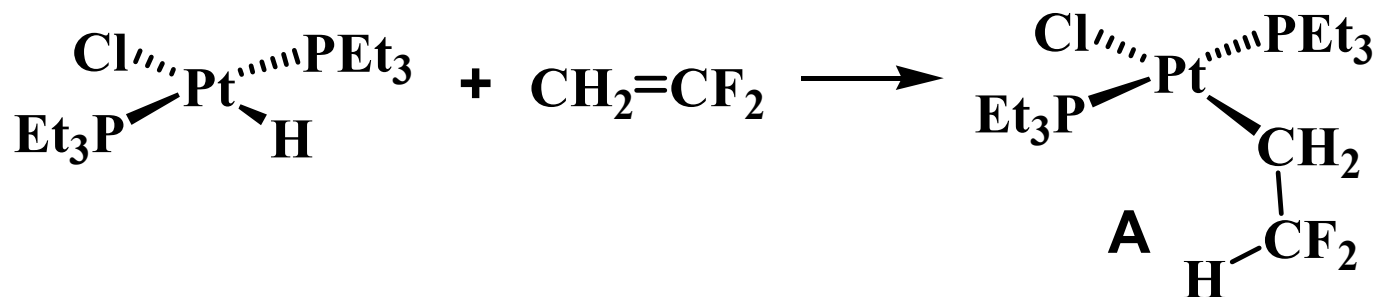
TS(A)



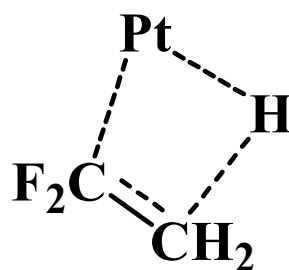
TS(B)

Answer: A

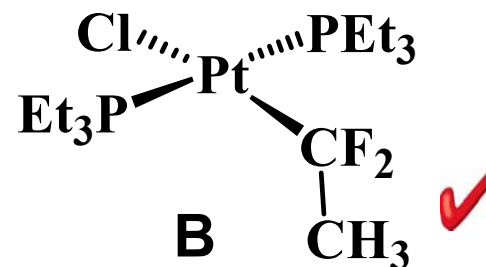
A is less steric repulsion



TS(A)



✓ TS(B)

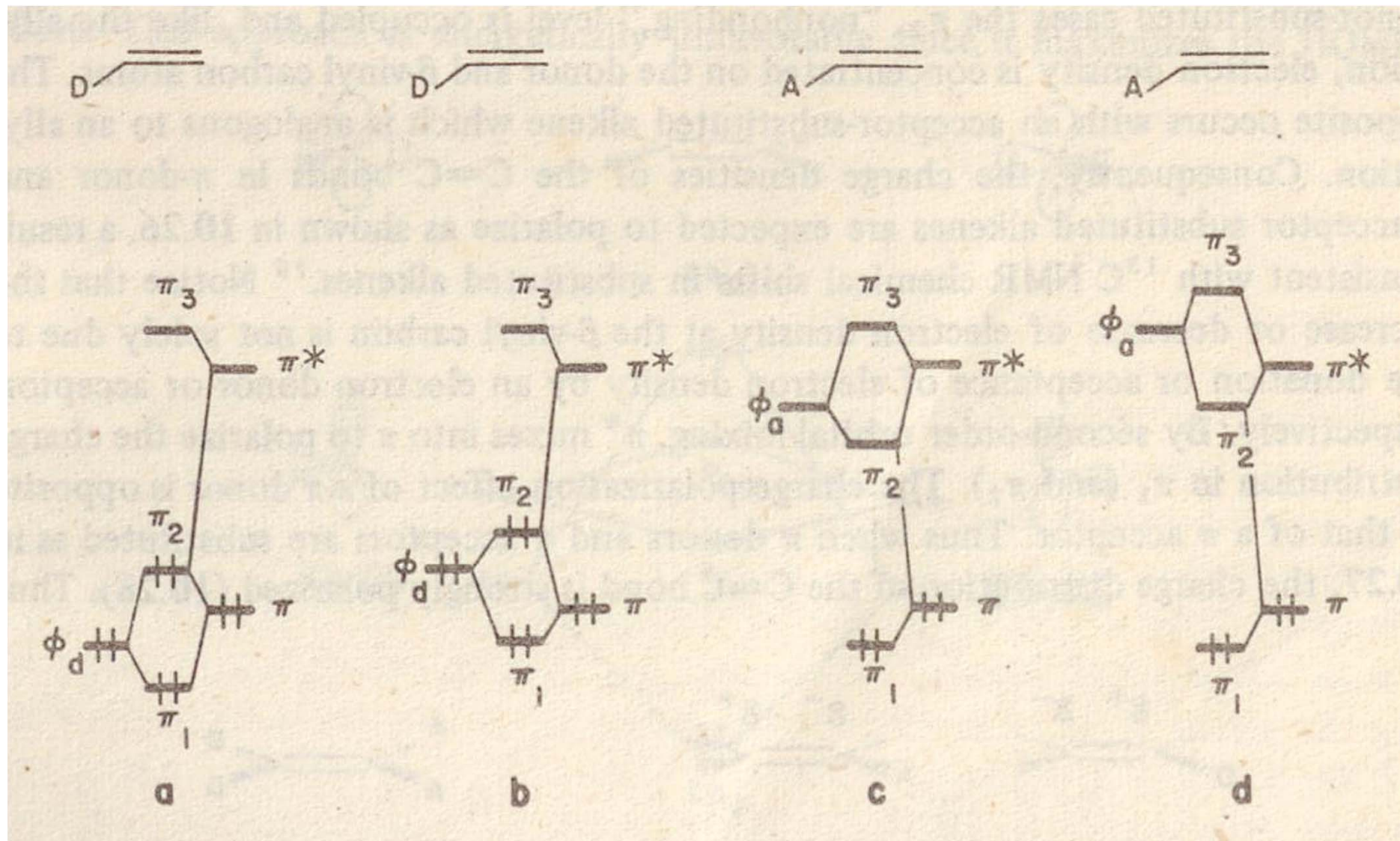


Which one is major ?

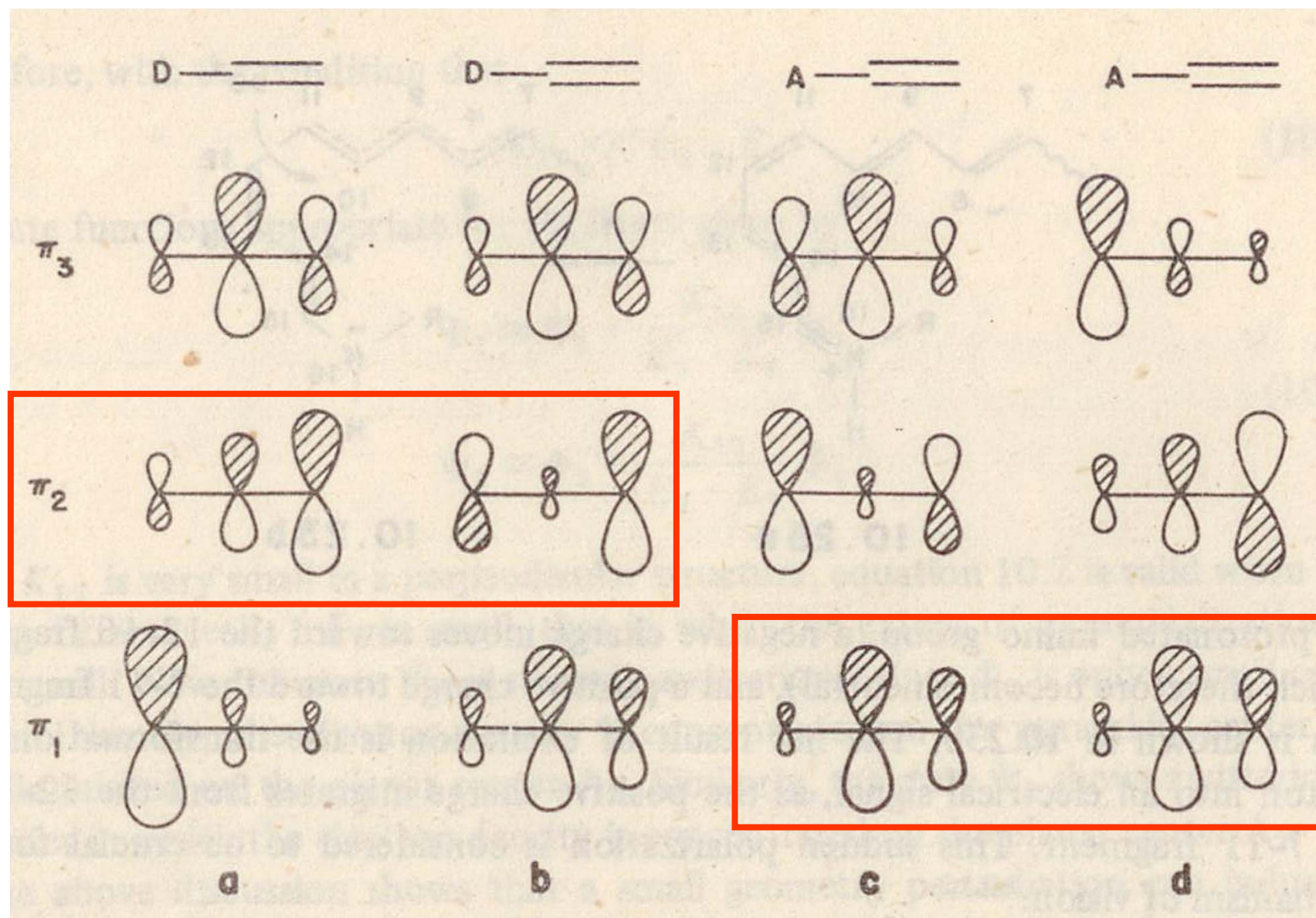
Answer: B

- Electronic reason, F is electron withdrawing group, CH₂ is more positive than CF₂.
- And also Pt-CF₂R bond is stronger than Pt-CH₂R

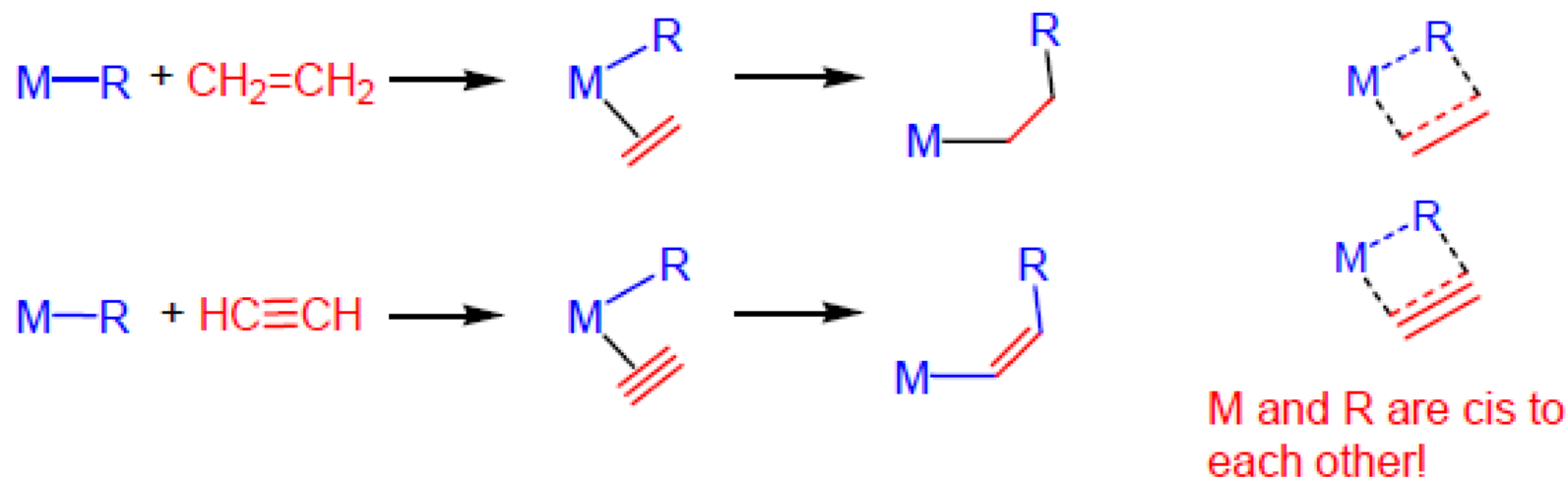
Substituent Effects on the C=C bond



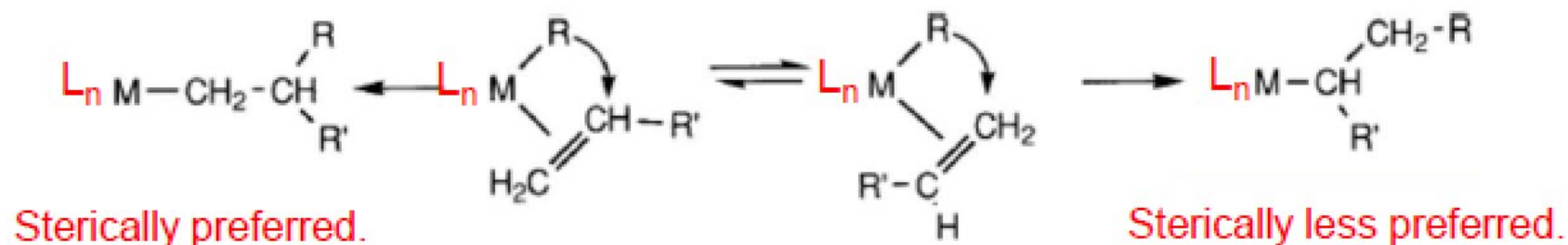
Substituent Effects on the C=C bond



- ✓ For the insertion reaction of alkynes, the stereochemistry is *syn* for the two substituents of $C\equiv C$.



- ✓ Regioselectivity for unsymmetrically substituted substrates

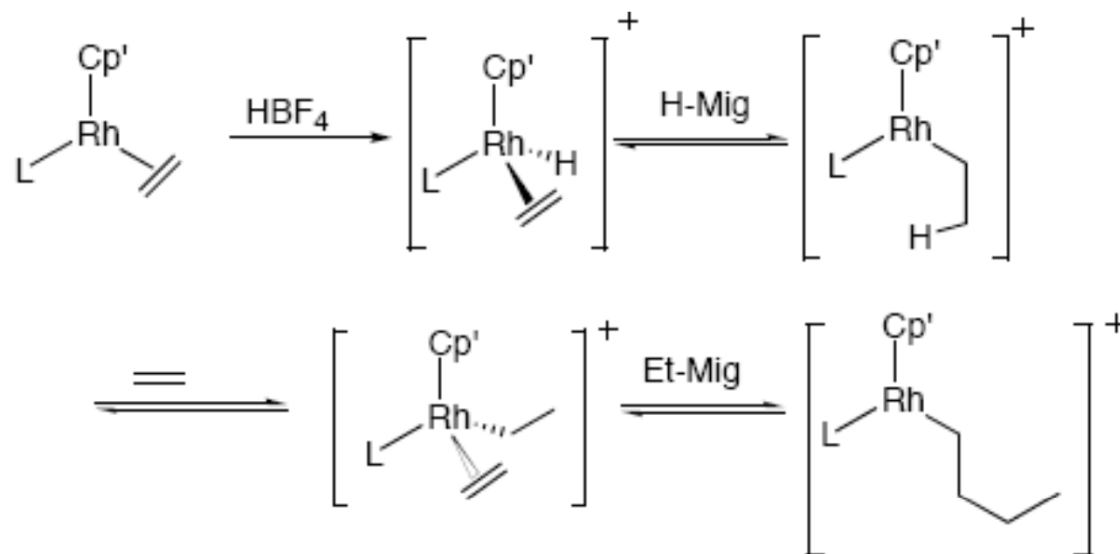


The preference may be reversed when R' is electron-withdrawing

Insertion into M-H versus M-R

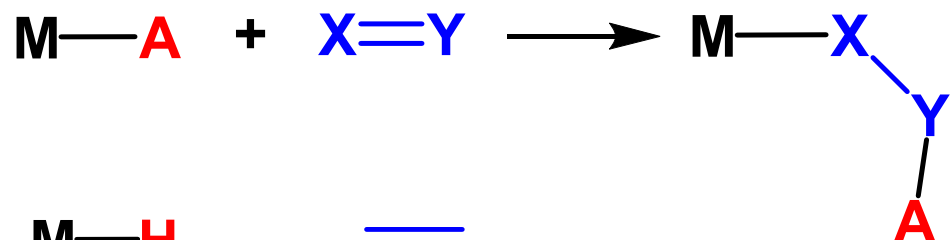
- Alkene insertion into M-R is thermodynamically favored over M-H insertion.
- However, migratory insertion into M-H bonds occurs at much higher rates than insertion into M-R bonds.

Therefore, the preference for M-H insertion must be due to kinetic factors, not thermodynamic factors. One example of the difference in activation energies was reported by Brookhart (*J. Am. Chem. Soc.*, **1992**, 114, 10394)



Cp'	L	$\Delta G^\ddagger$ H-Mig (kcal/mol)	$\Delta G^\ddagger$ Et-Mig (kcal/mol)
Cp*	P(OMe) ₃	12.2	22.4
Cp*	PMe ₃	12.1	23.4
Cp	P(OMe) ₃	15.0	23.4
Cp	PMe ₃	15	24.7

3). Related insertion reactions. $X \equiv Y$

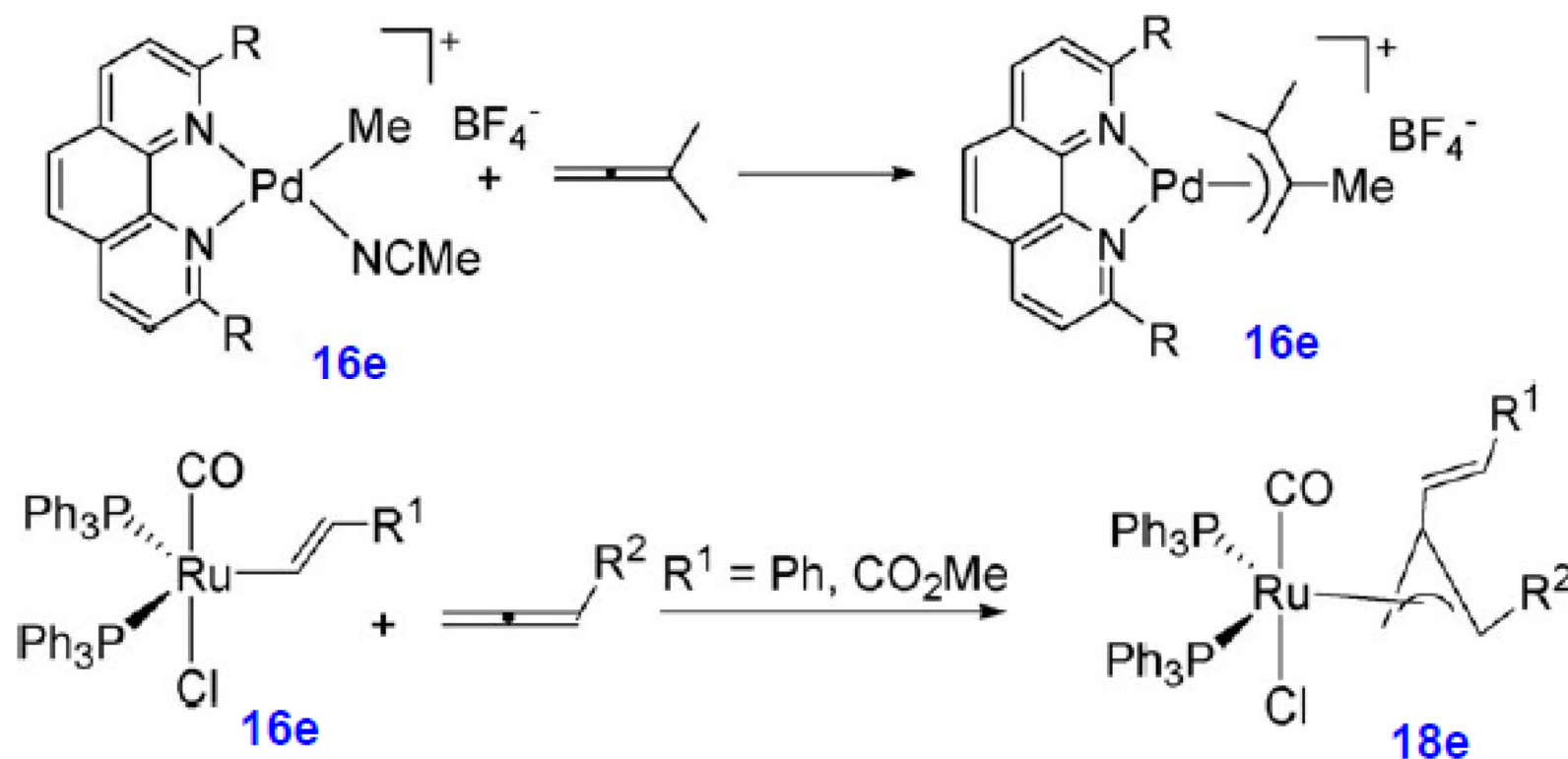
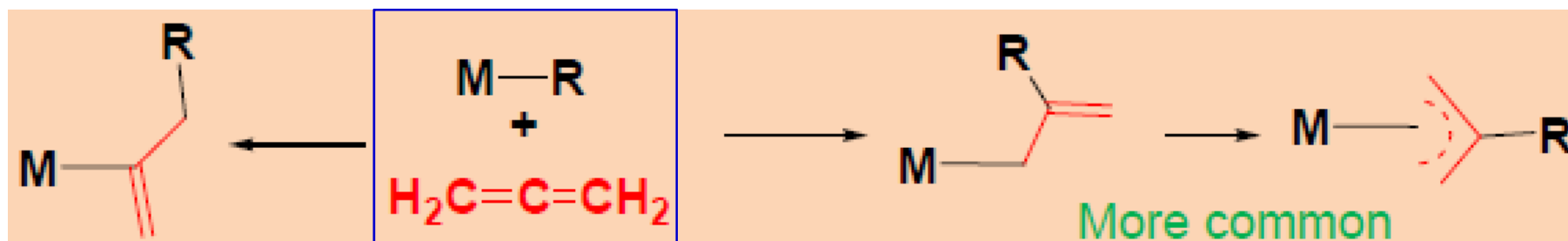


✓ In general, the reactions are mechanistically similar to those of alkenes.

✓ Substrates with polar bonds (e.g. $R_2C=O$, CO_2) may also react through other routes.

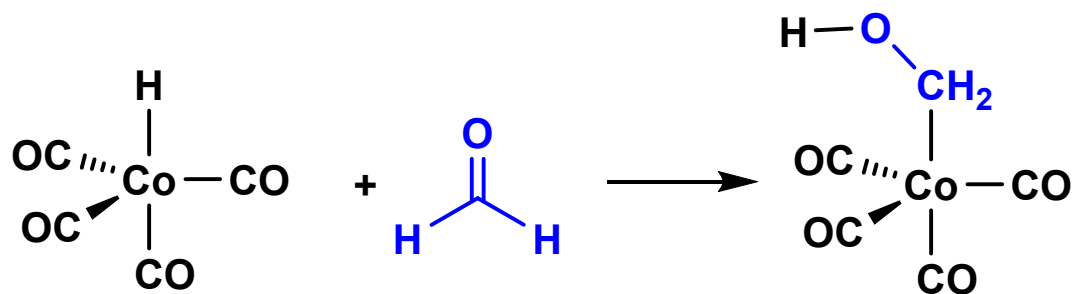
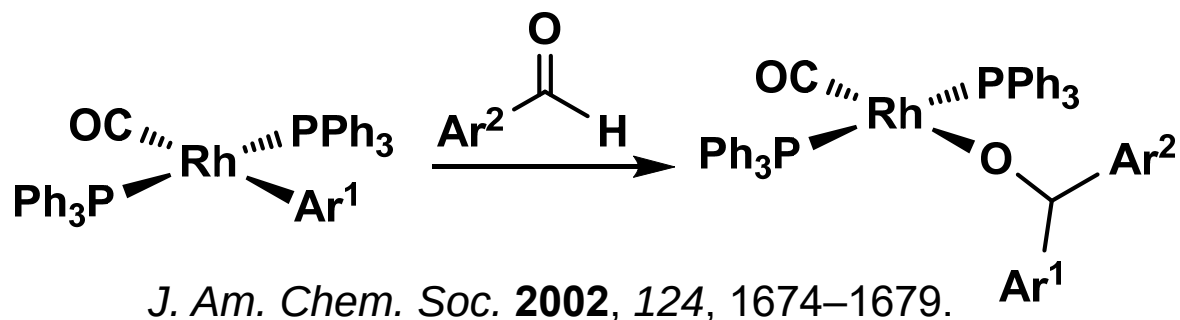
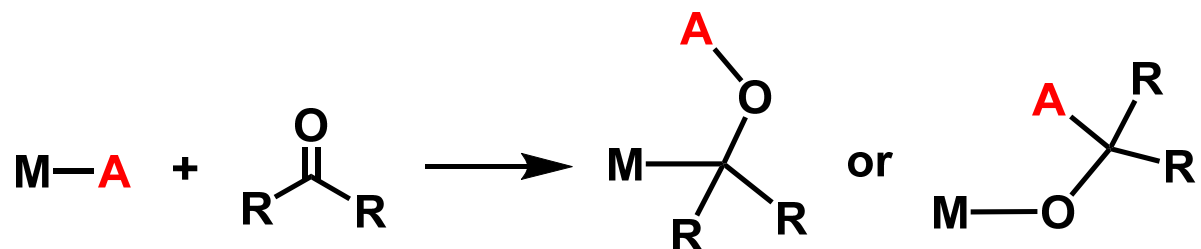
3). Related insertion reactions. *Examples.*

✓ *Insertion reactions of allenes*



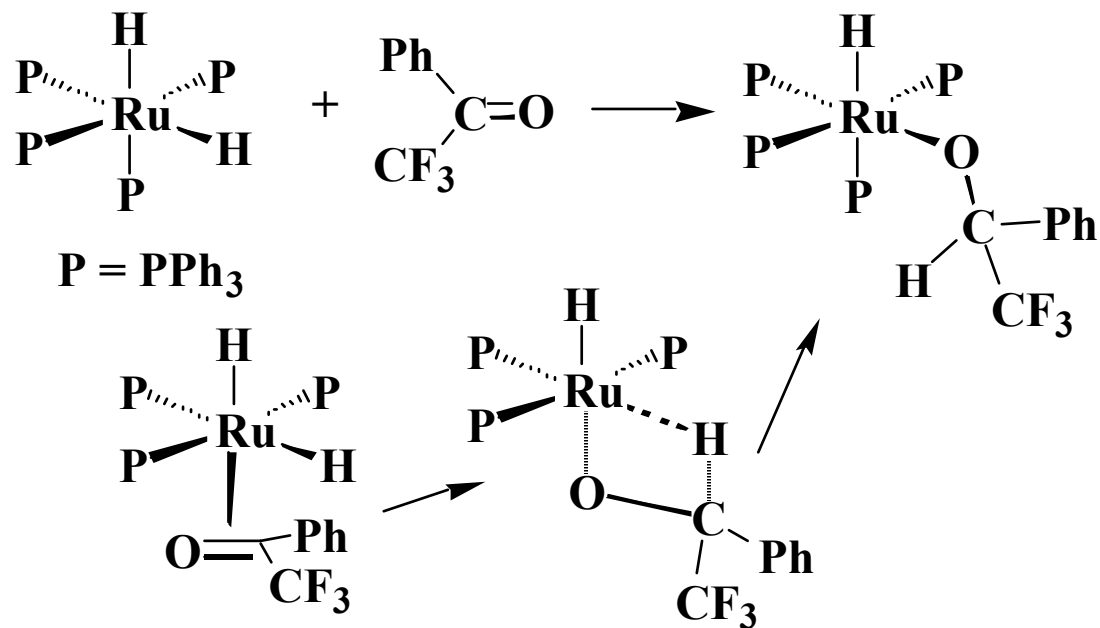
3). Related insertion reactions. *Examples.*

✓ Insertions of ketone and aldehydes, like $\text{CH}_2=\text{CH}_2$



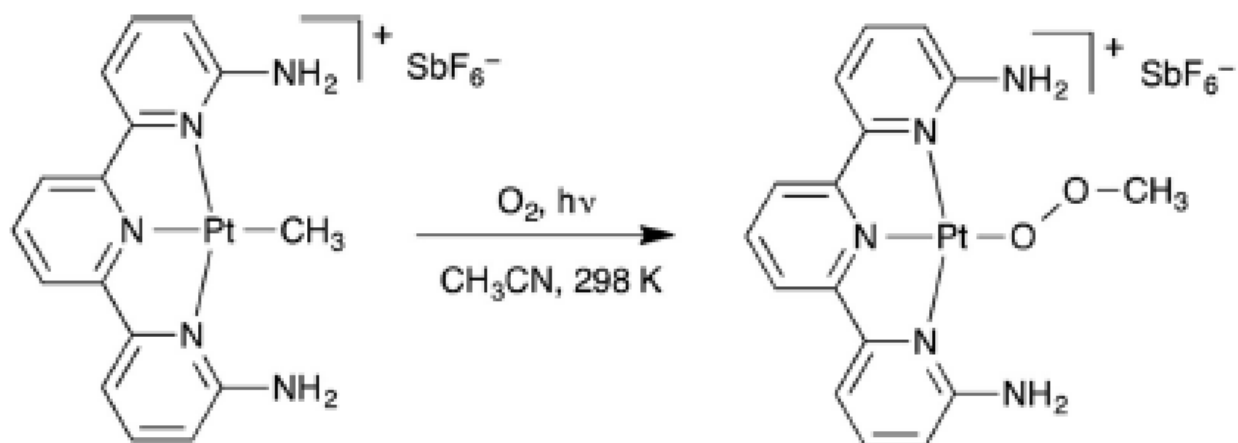
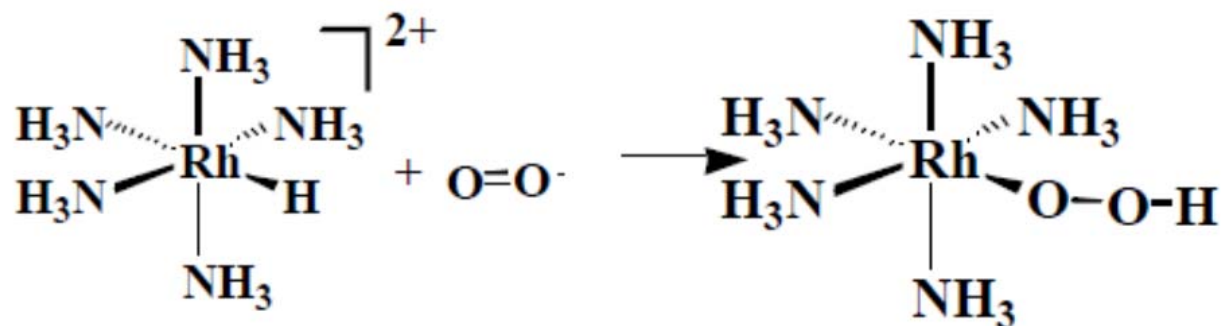
3). Related insertion reactions. *Examples.*

✓ Insertions of ketone and aldehydes, like $\text{CH}_2=\text{CH}_2$



3). Related insertion reactions. *Examples.*

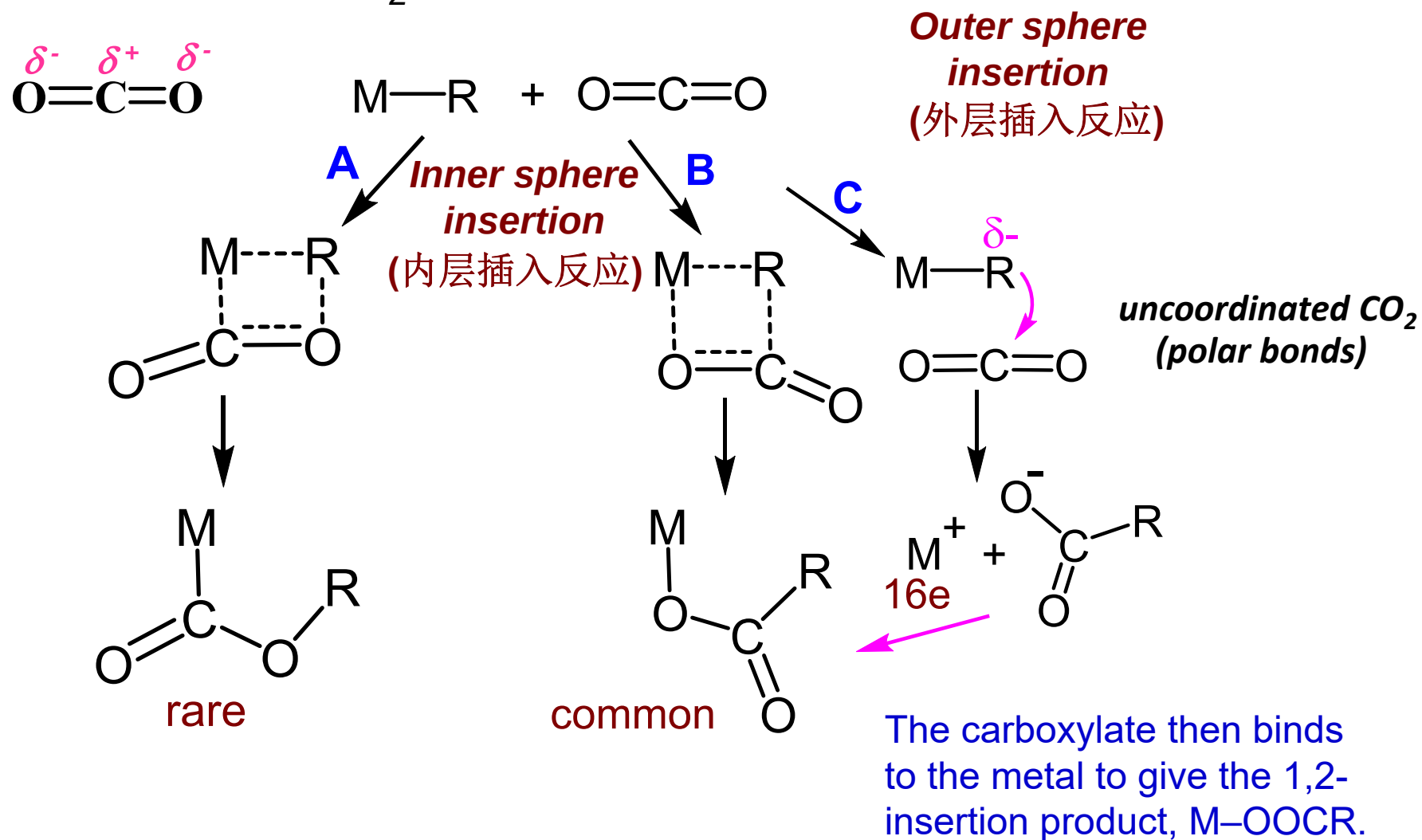
✓ Insertions of O_2 , like $CH_2=CH_2$



J. Am. Chem. Soc. **2014**, 136, 14089–14099.

3). Related insertion reactions.

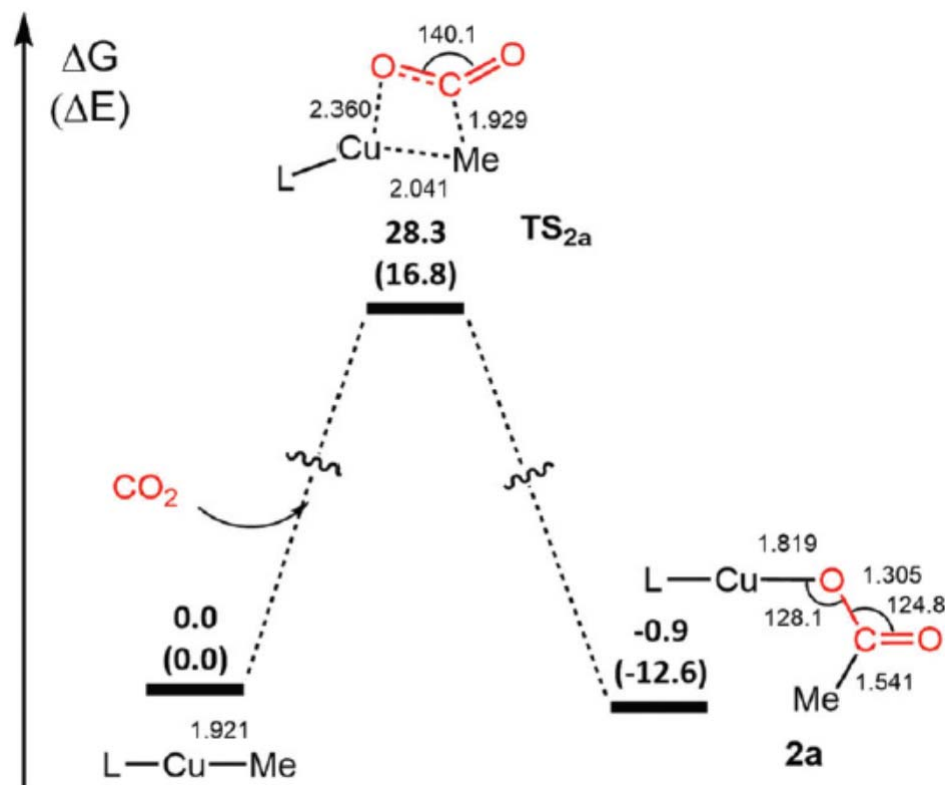
✓ Insertions of CO_2



3). Related insertion reactions.

✓ Insertions of CO_2

Example 1.



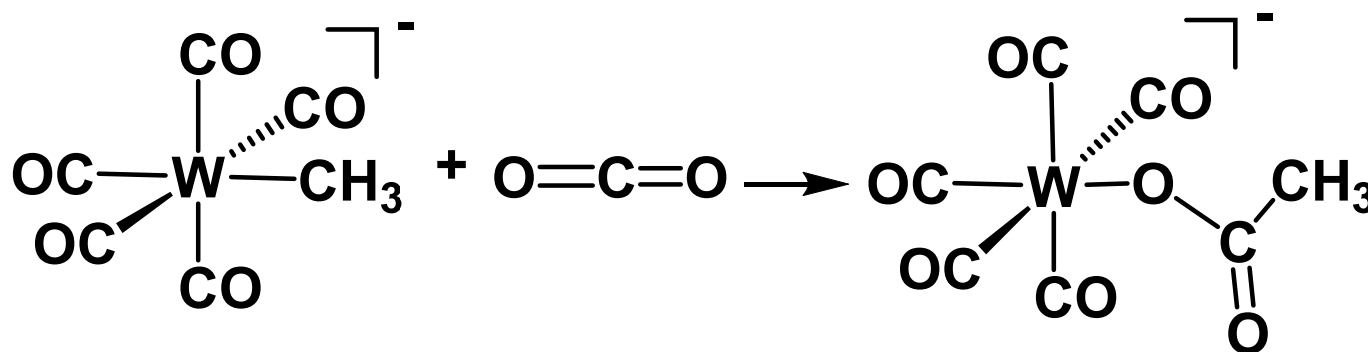
Zhenyang Lin, *et. al*, *New J. Chem.*, **2021**, 45, 15022.

Outer Sphere Insertions (外层插入反应)

In some cases, the A=B bond does not need to coordinate to the metal prior to insertion and can undergo the reaction with an 18e complex.

The weakly binding ligand, CO₂, can insert into an M-H or M-R bond in this way.

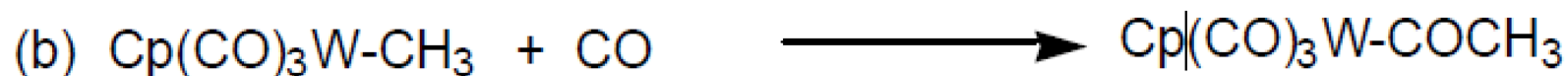
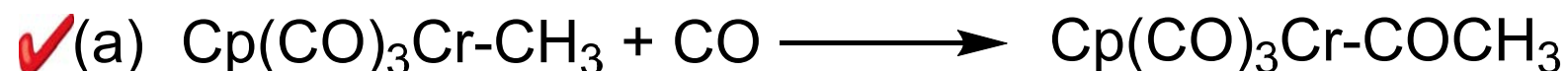
Example 2.



18e, direct attack of uncoordinated CO₂.

Review questions

1. Which of the following reactions would you expect to have the highest reaction rate?

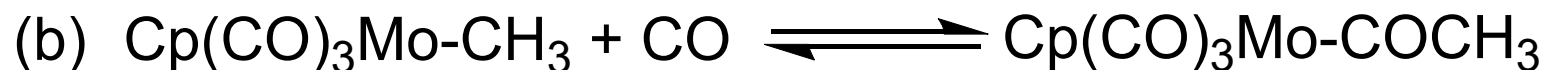
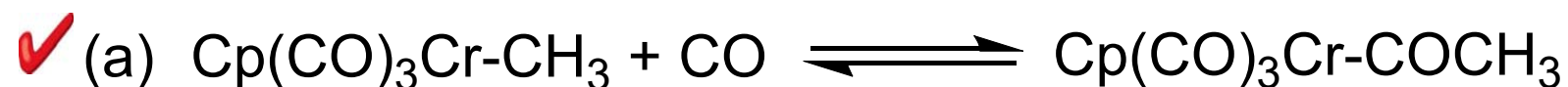


Bond strength

Cr-CH₃ bond is weak

Review questions

2. Which of the following reactions would you expect to have the largest equilibrium constant K_{eq} ?

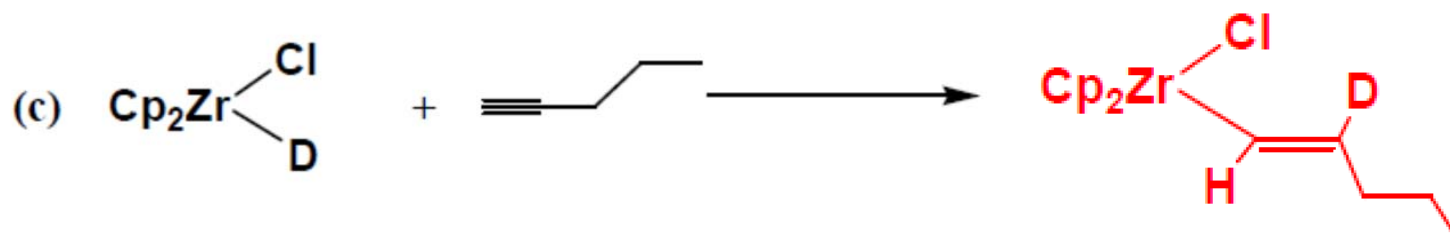
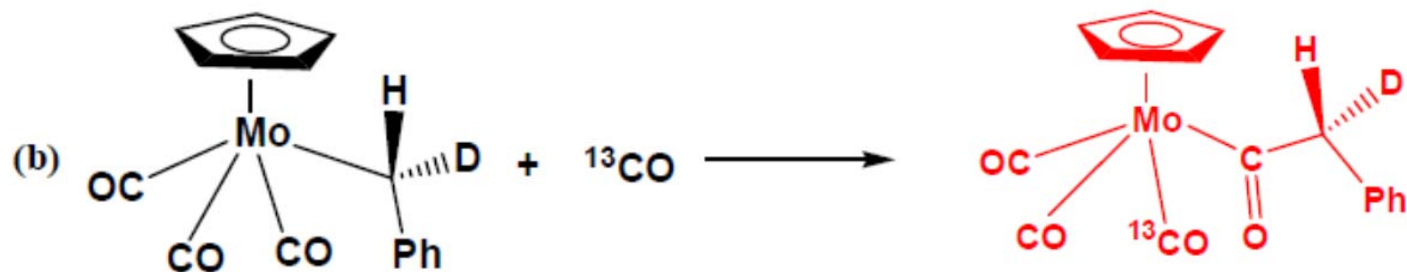
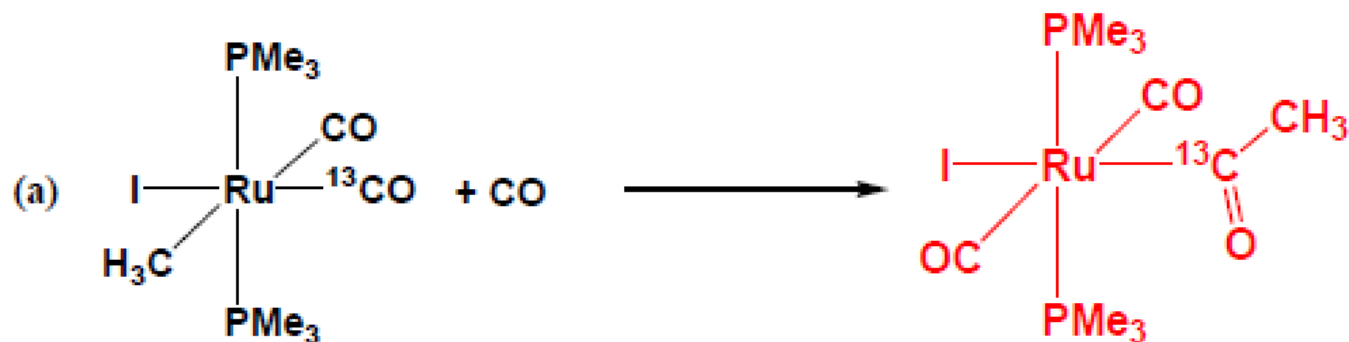


Bond strength

Cr-CH₃ bond is weak

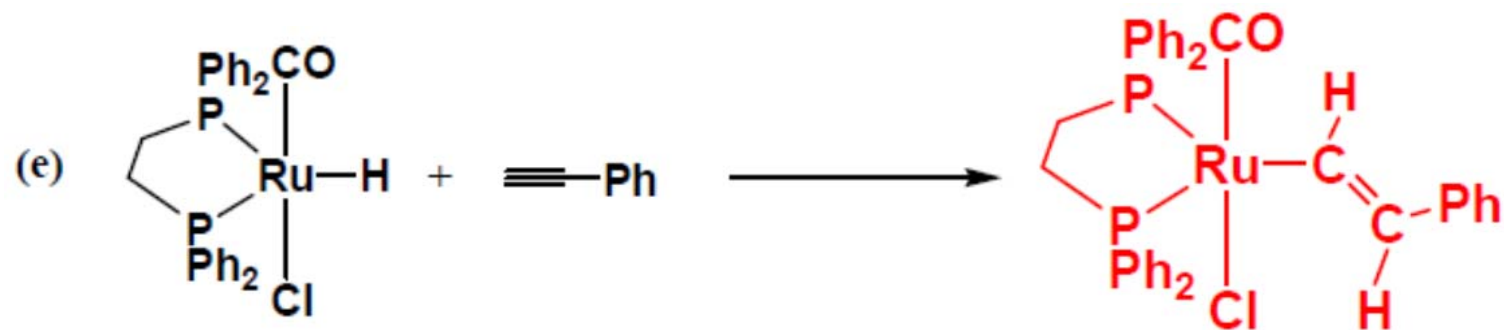
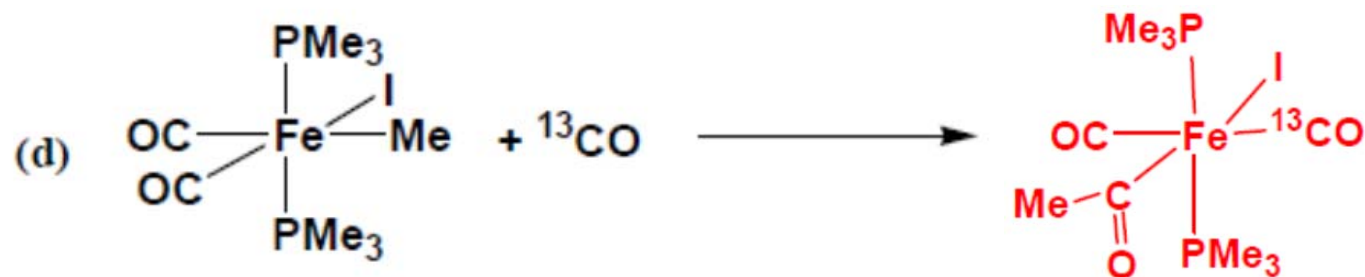
Review questions

3. Predict the products?



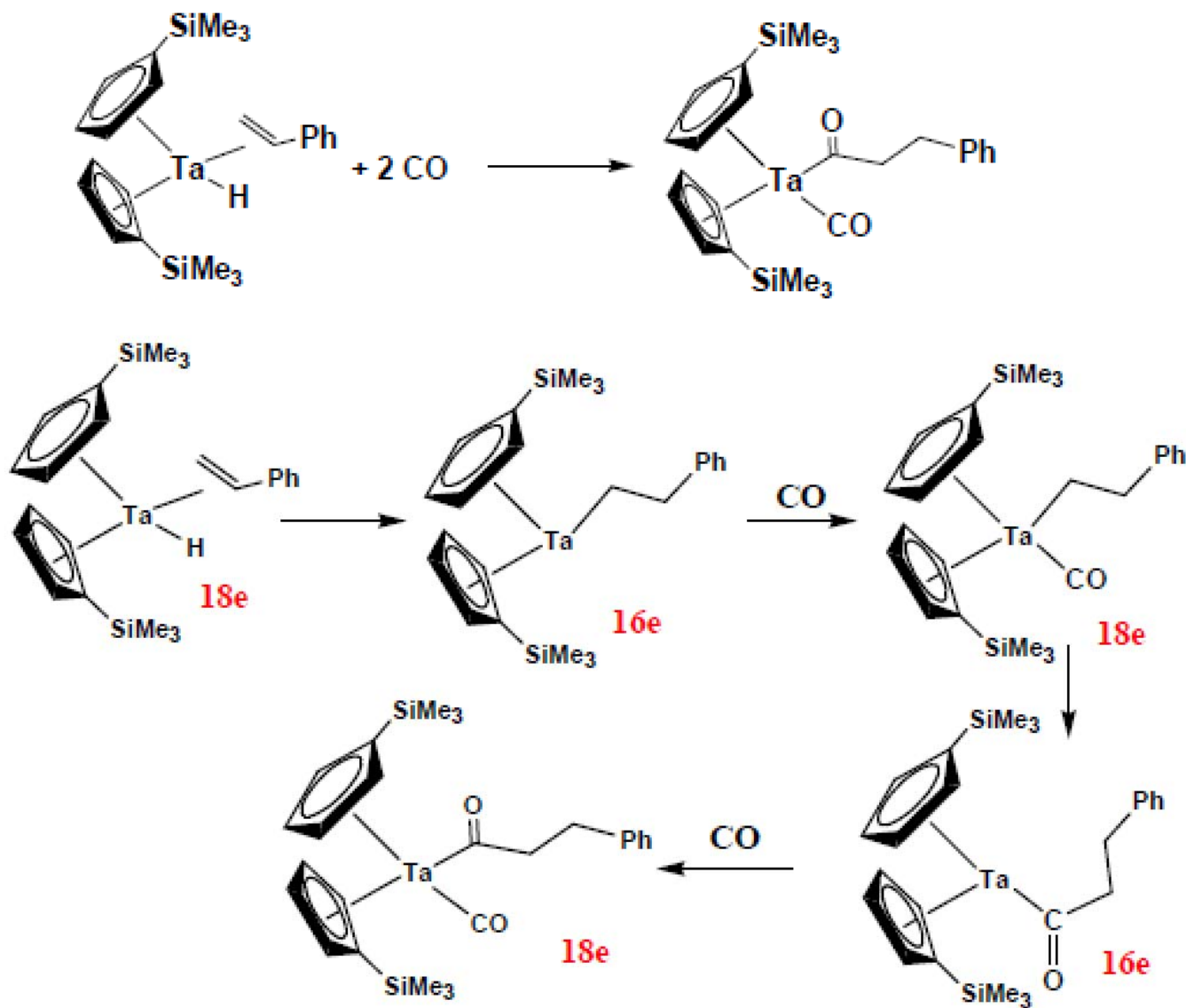
Review questions

3. Predict the products?



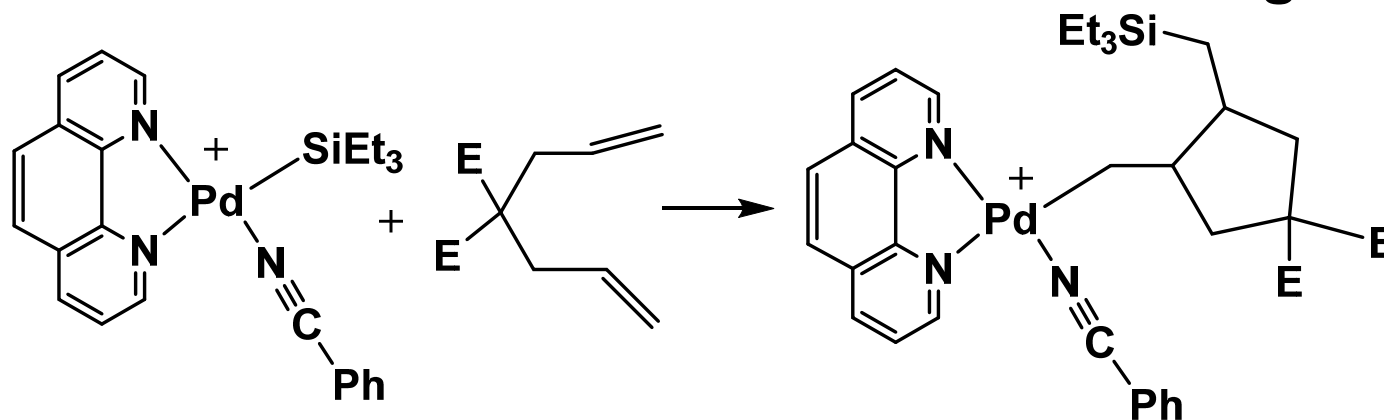
Review questions

4. Suggest a reasonable mechanism for the following reaction.



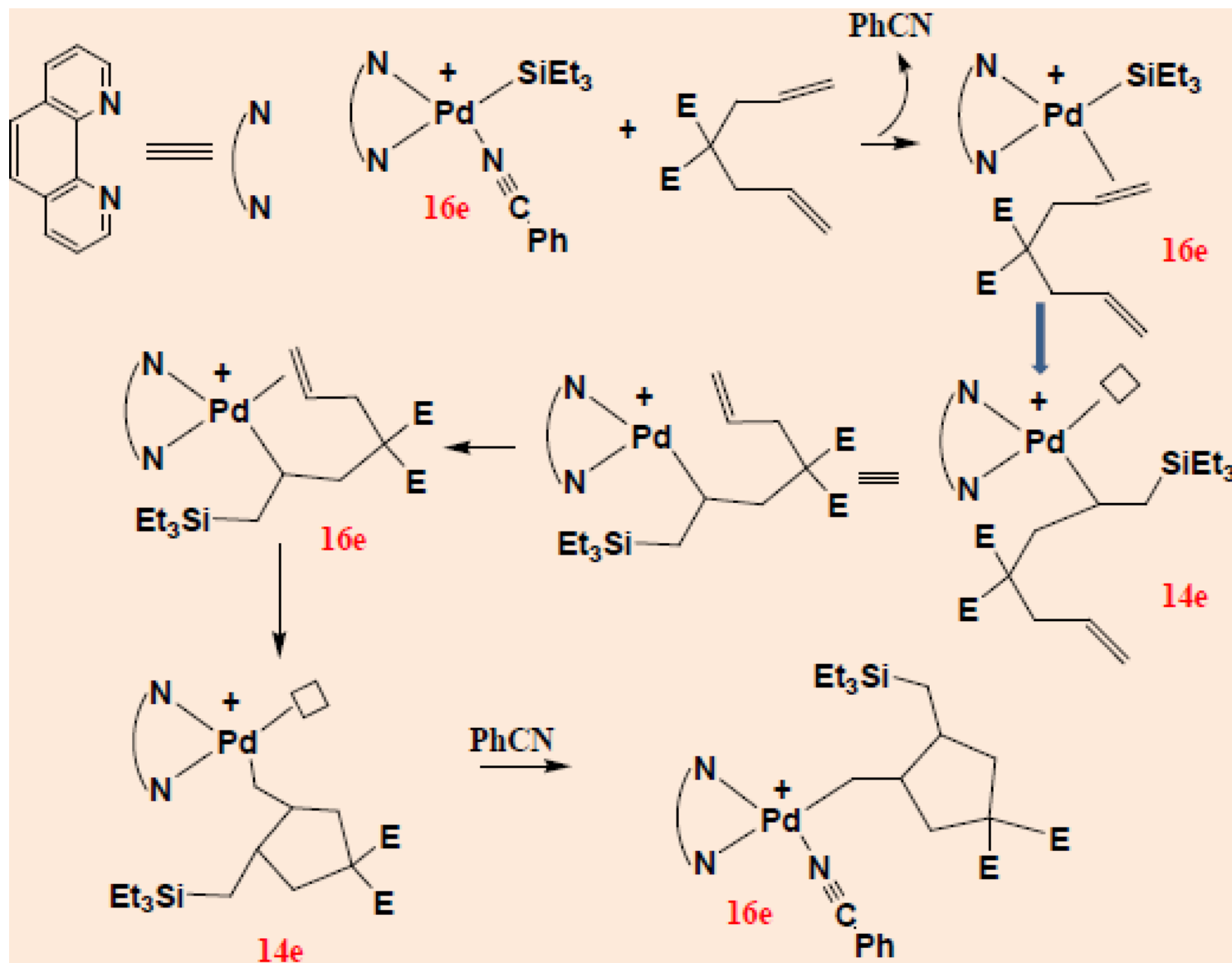
Review questions

5. Suggest a reasonable mechanism for the following reaction.



Review questions

5. Suggest a reasonable mechanism for the following reaction.

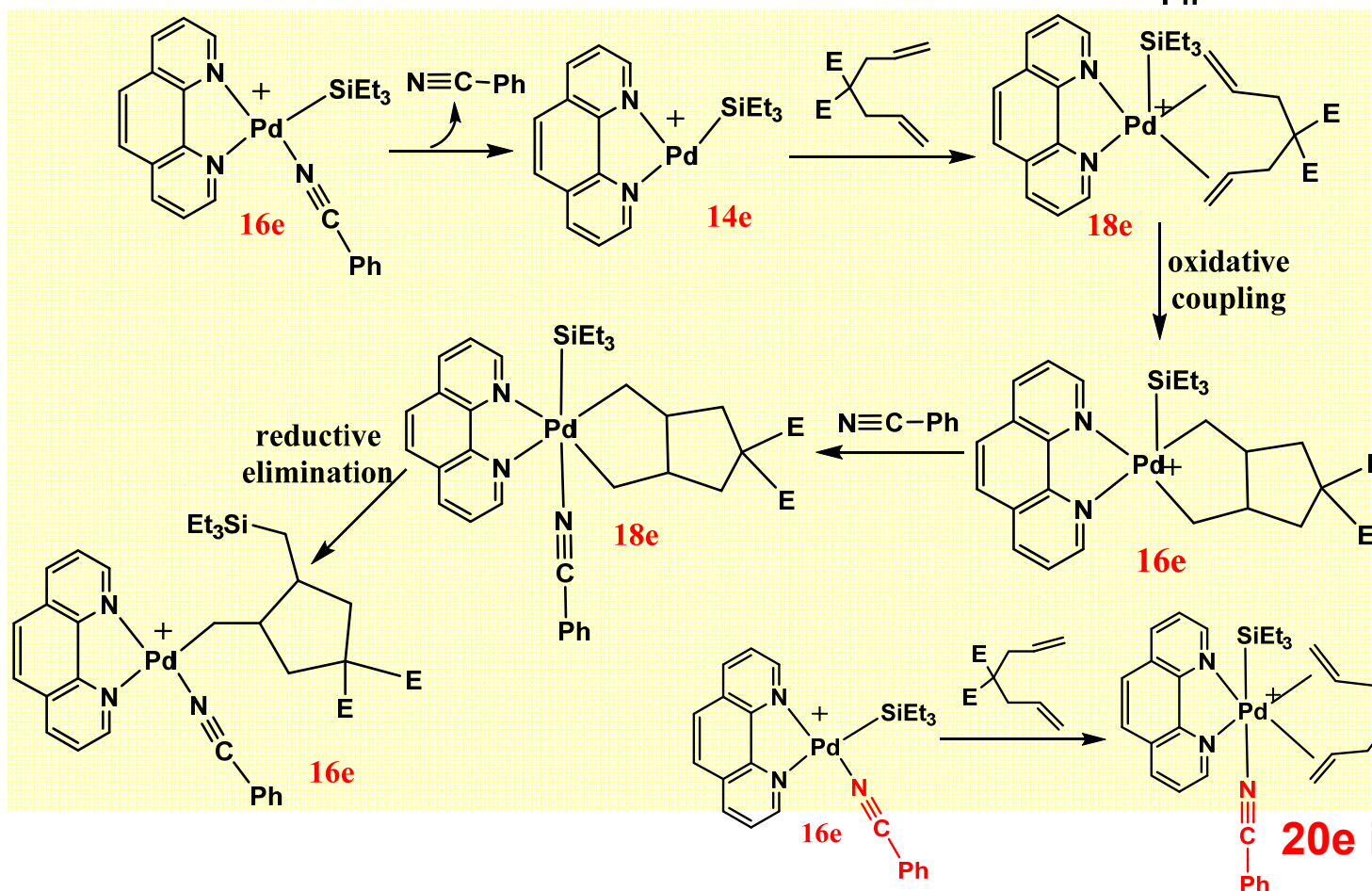
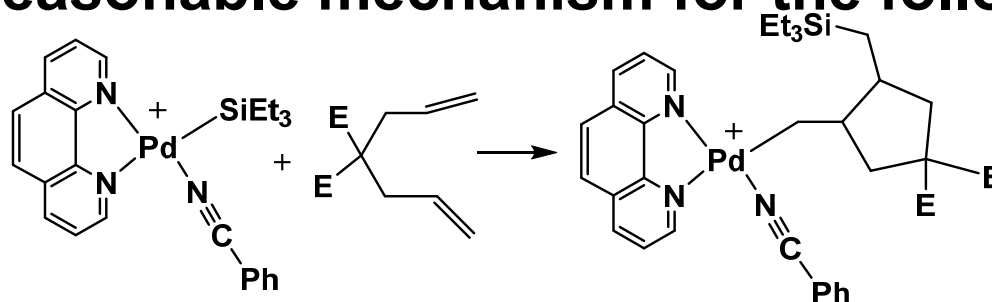


Prefer Pd(II) 16e square-planar intermediate.



Review questions

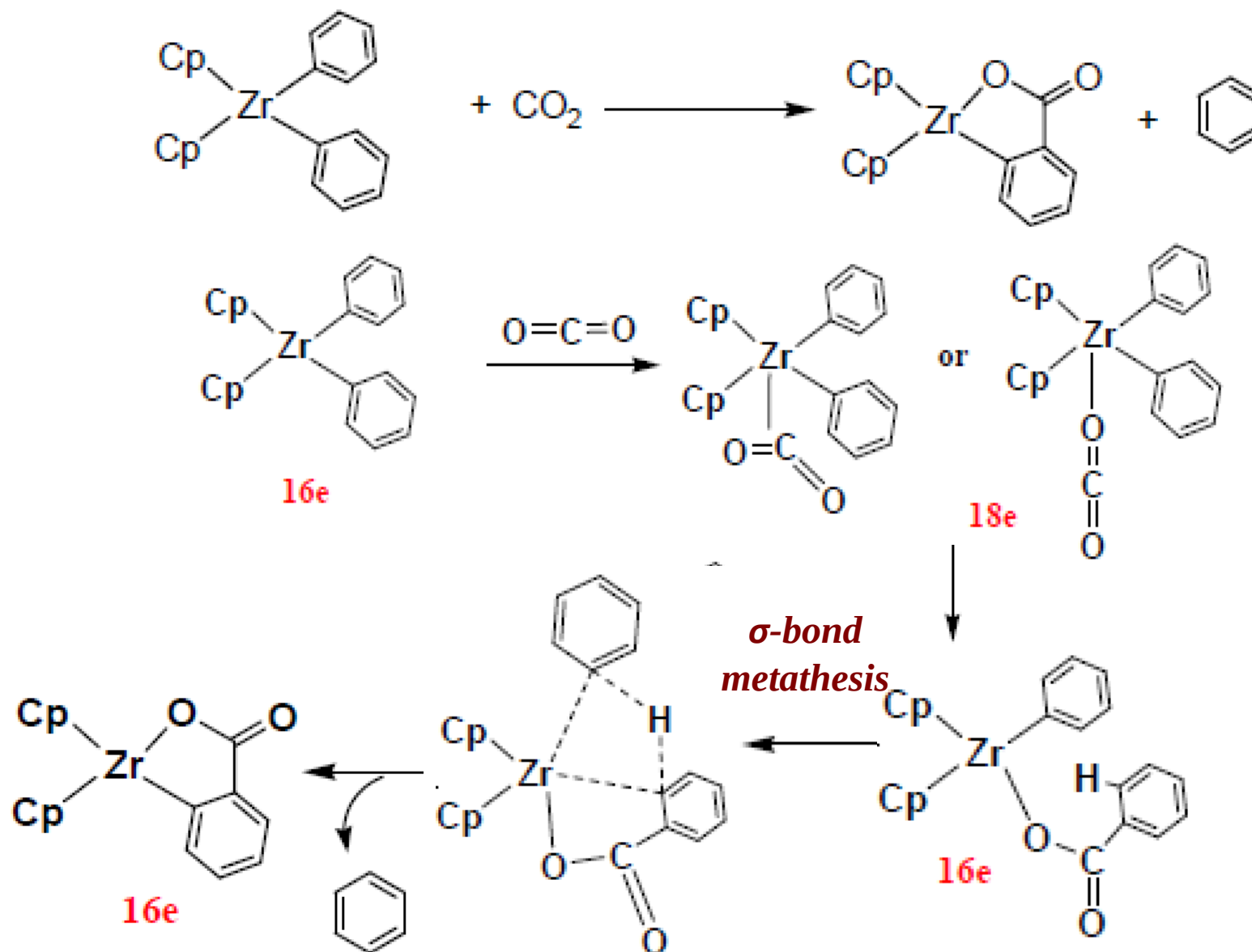
5. Suggest a reasonable mechanism for the following reaction.



5-coordinate
18e Pd(II)
intermediate
may also have
a certain
degree of
rationality. ✓

Review questions

6. Suggest a reasonable mechanism for the following reaction.



D. α , β , γ and δ elimination reactions.

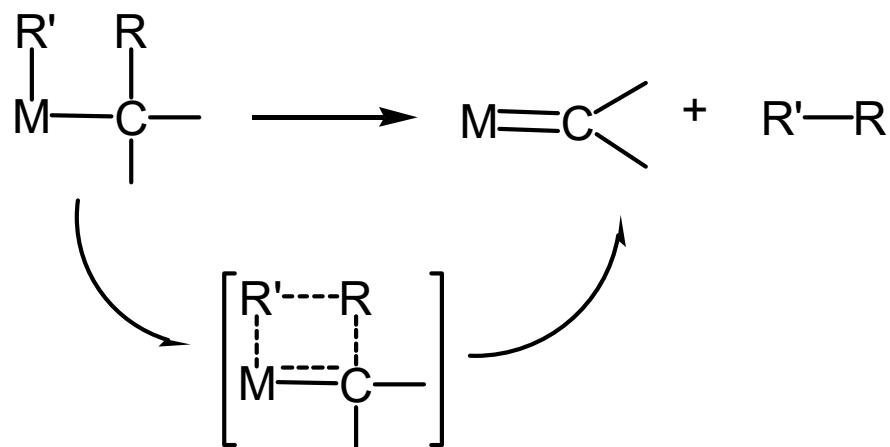
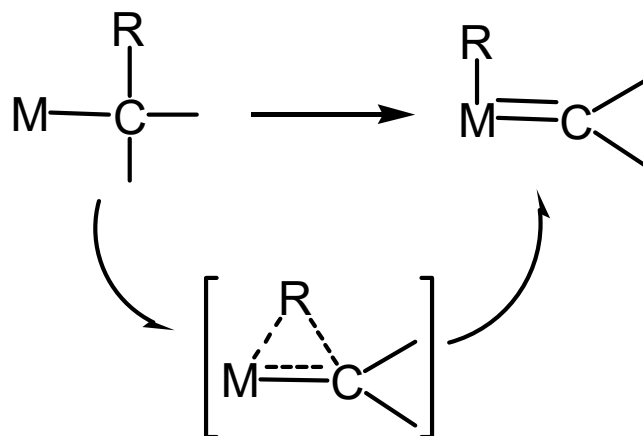
1). α elimination

2). β elimination

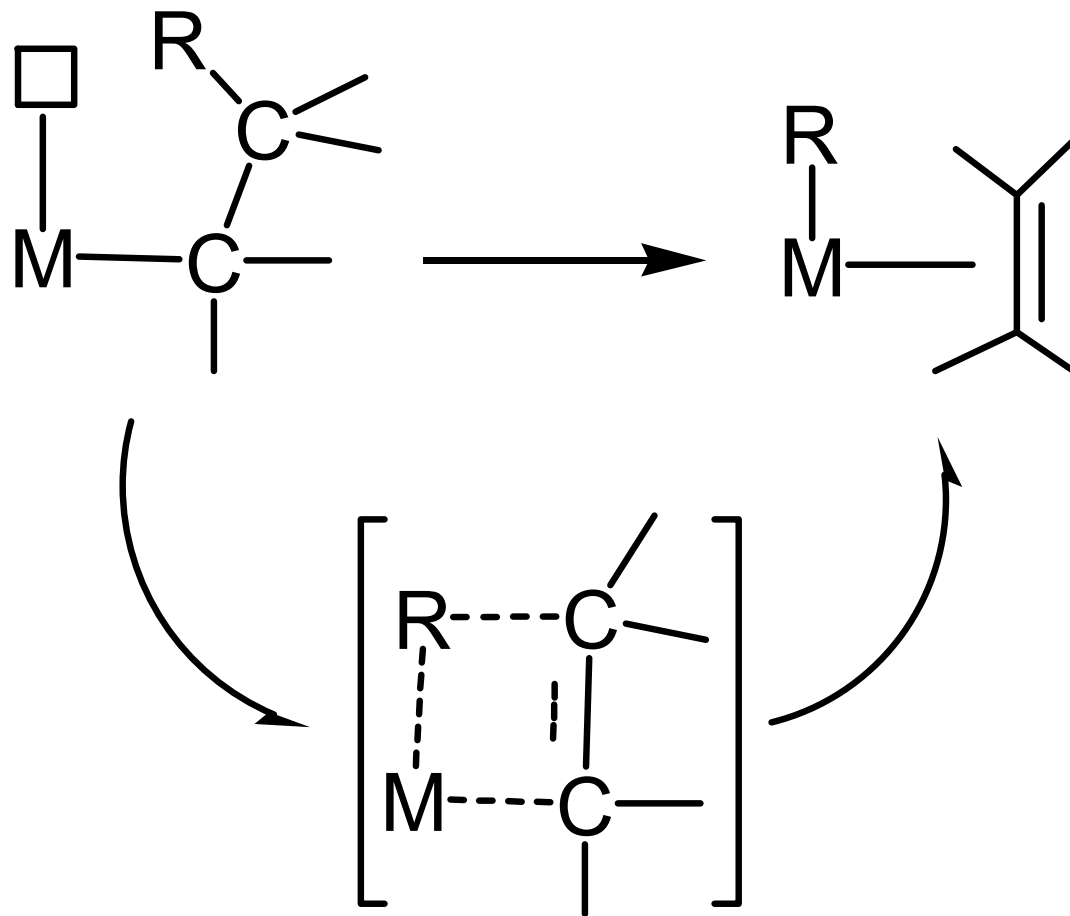
3). γ elimination

4). δ elimination

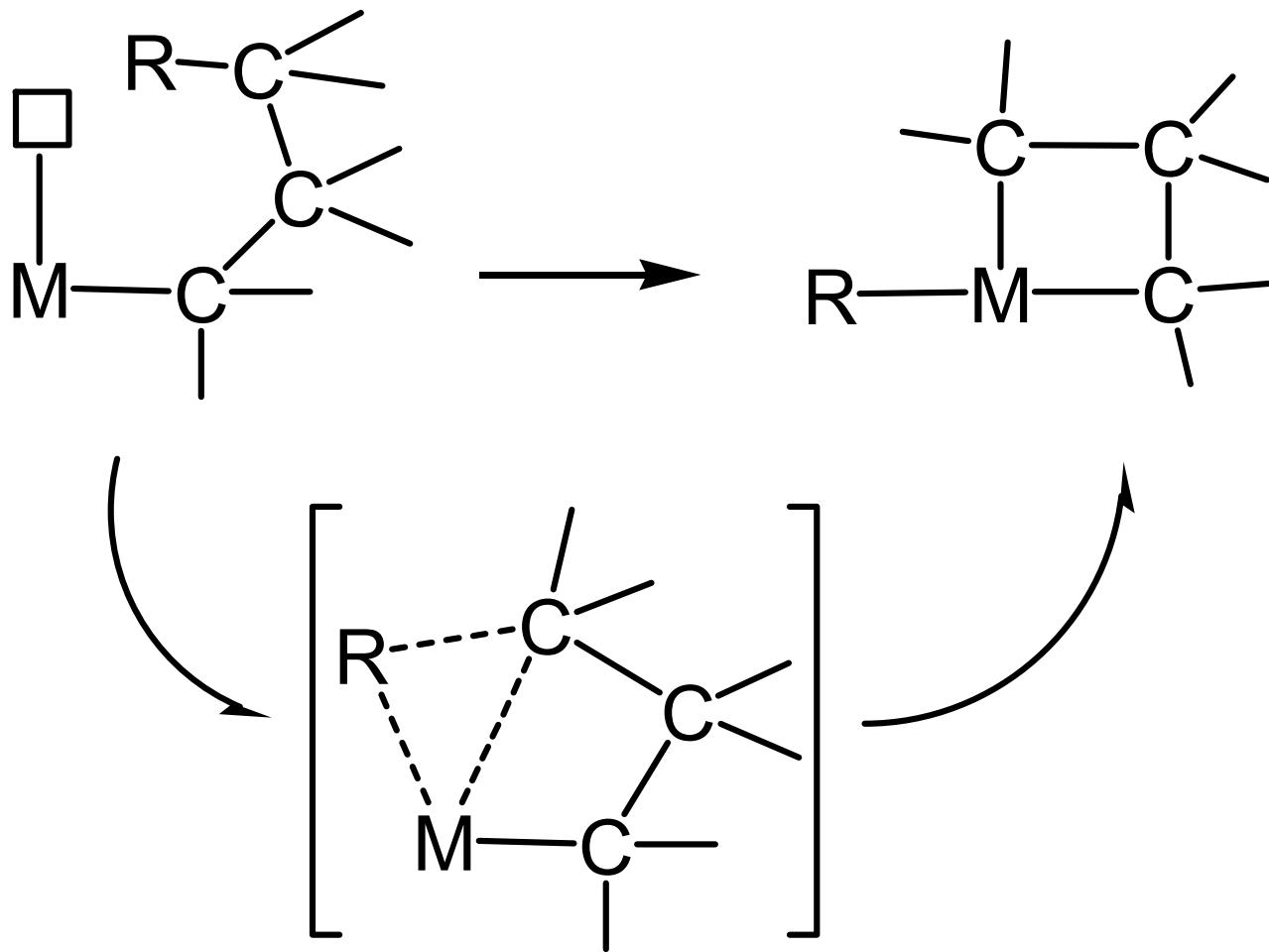
1). α -elimination



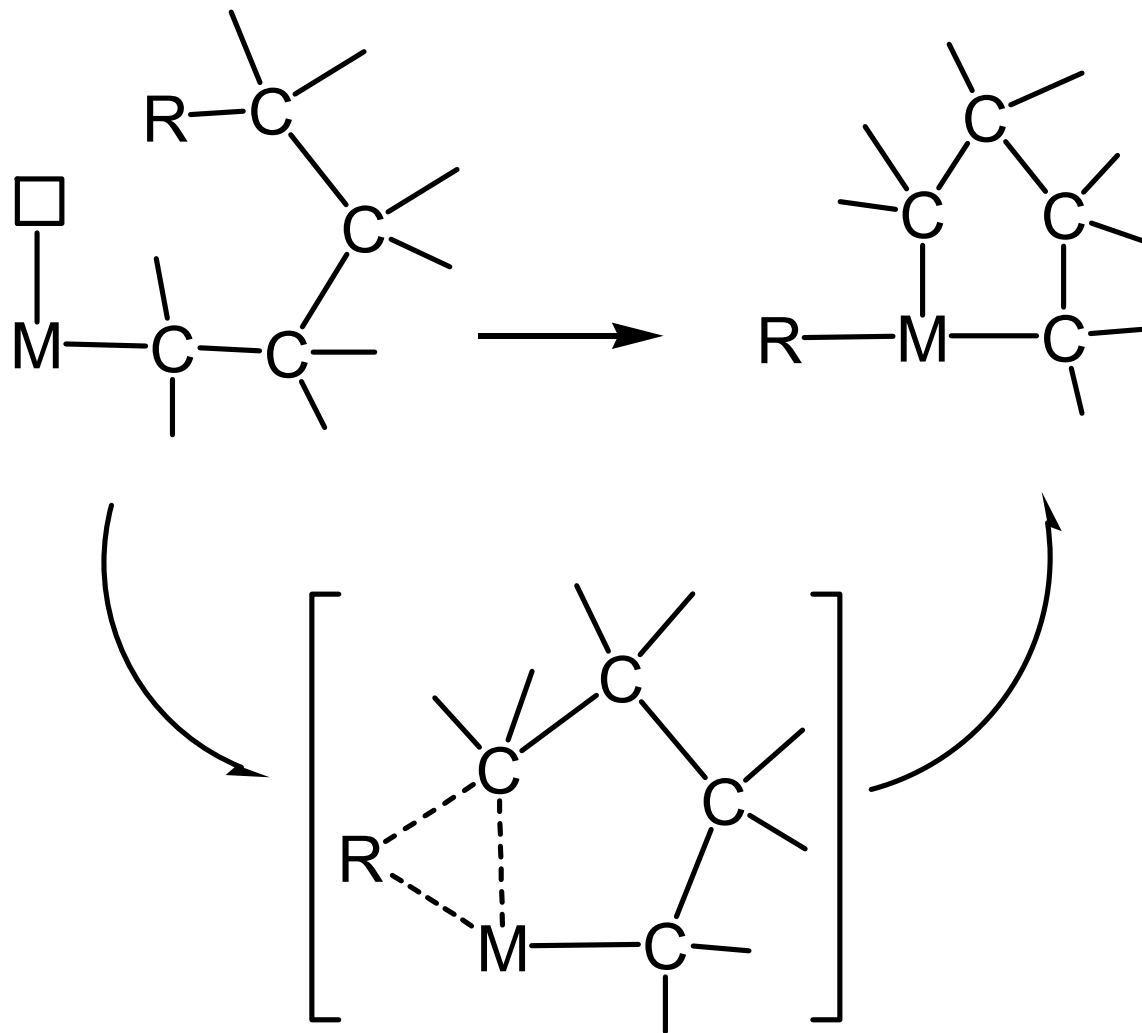
2). β -elimination



3). γ -elimination

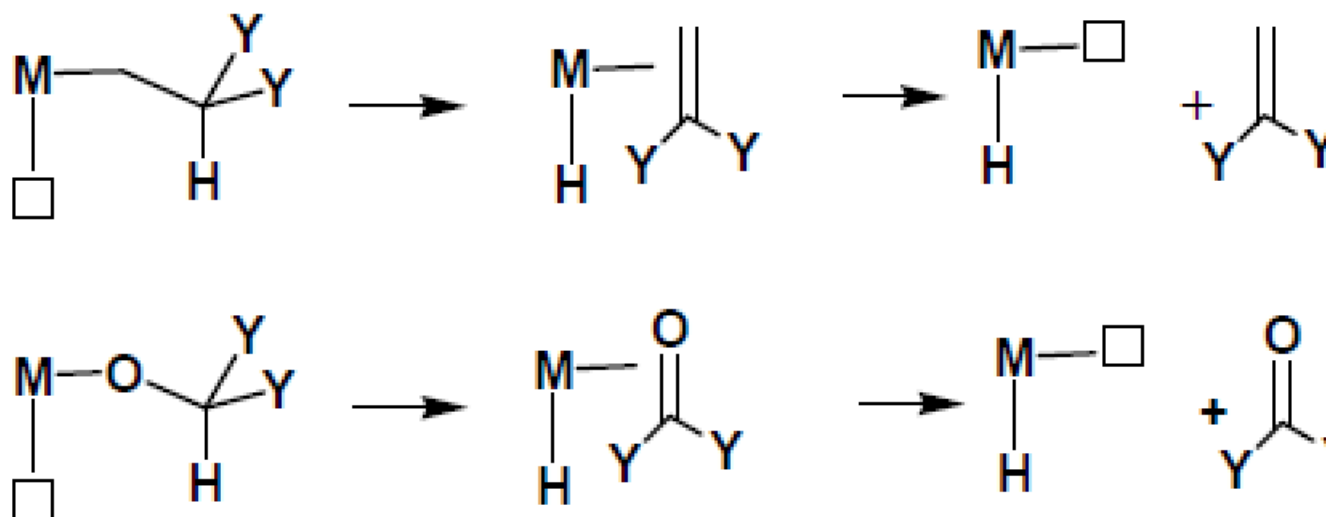


4). δ -elimination



β -Elimination

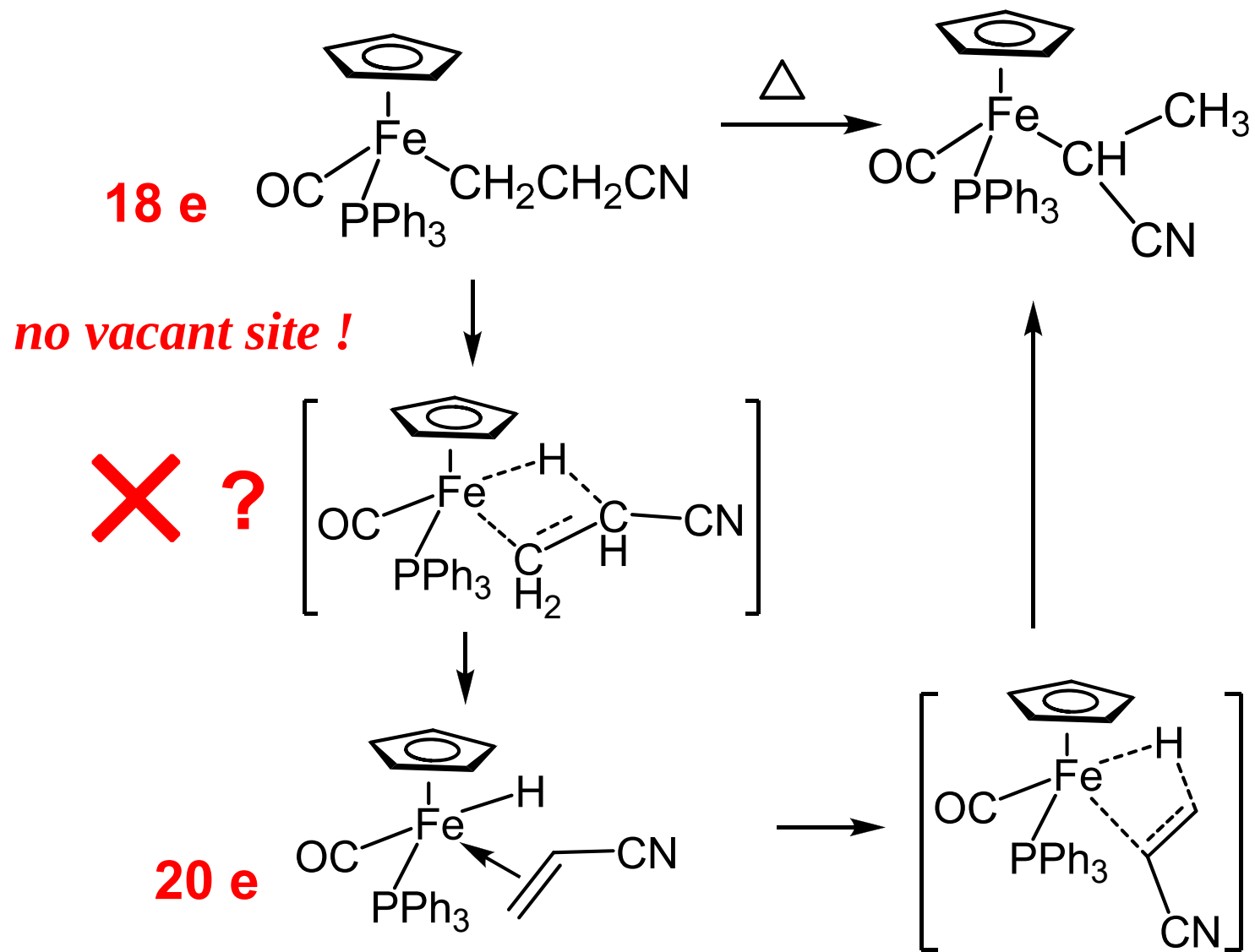
- β -H elimination reactions are very common reactions. E.g.



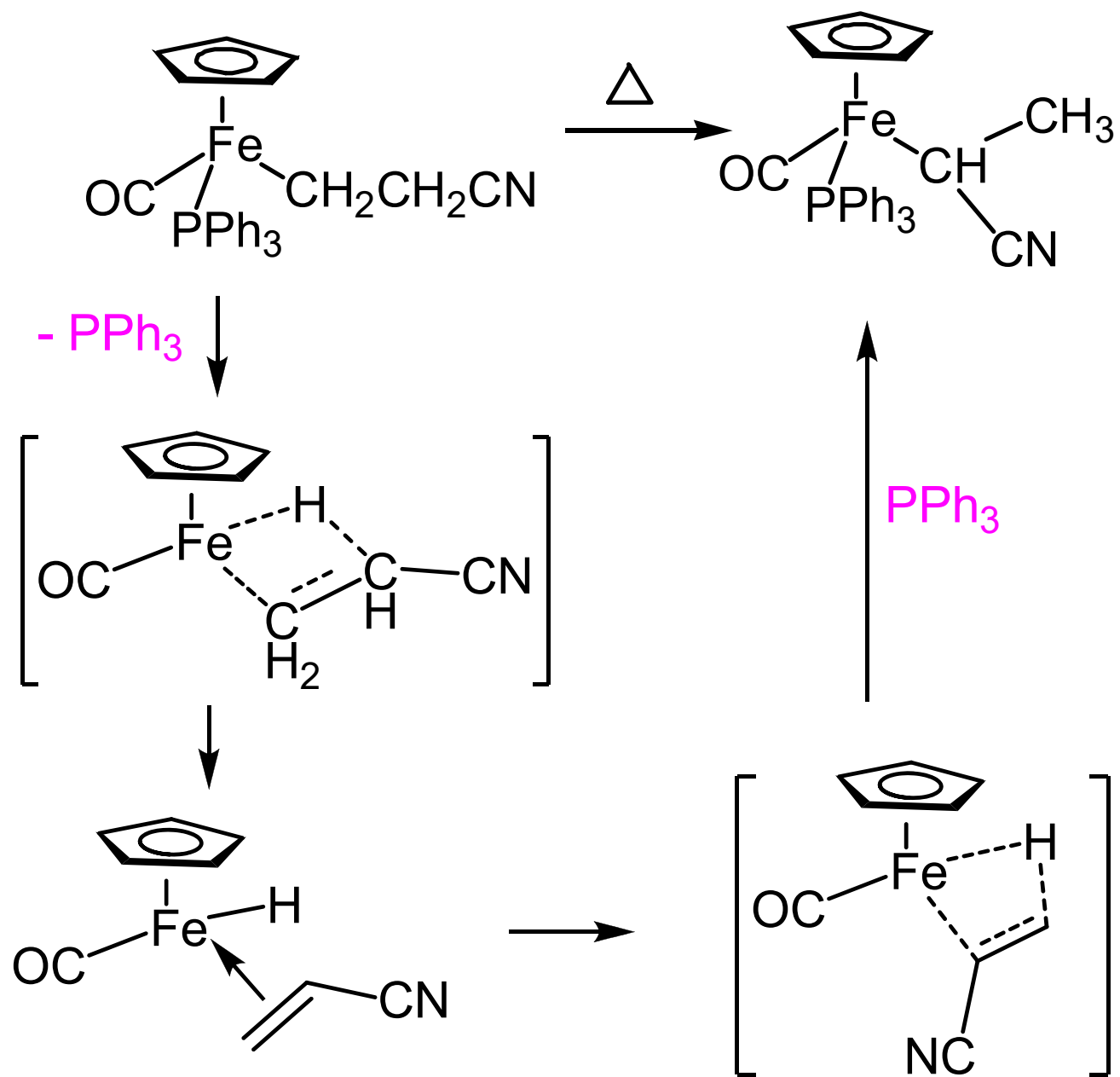
- The reactions can be irreversible or reversible.
- ✓ Irreversible reactions have often been used to make M-H complexes.
 - ✓ Reversible reactions are often involved in organometallic reactions and catalysis.

Practice Problems

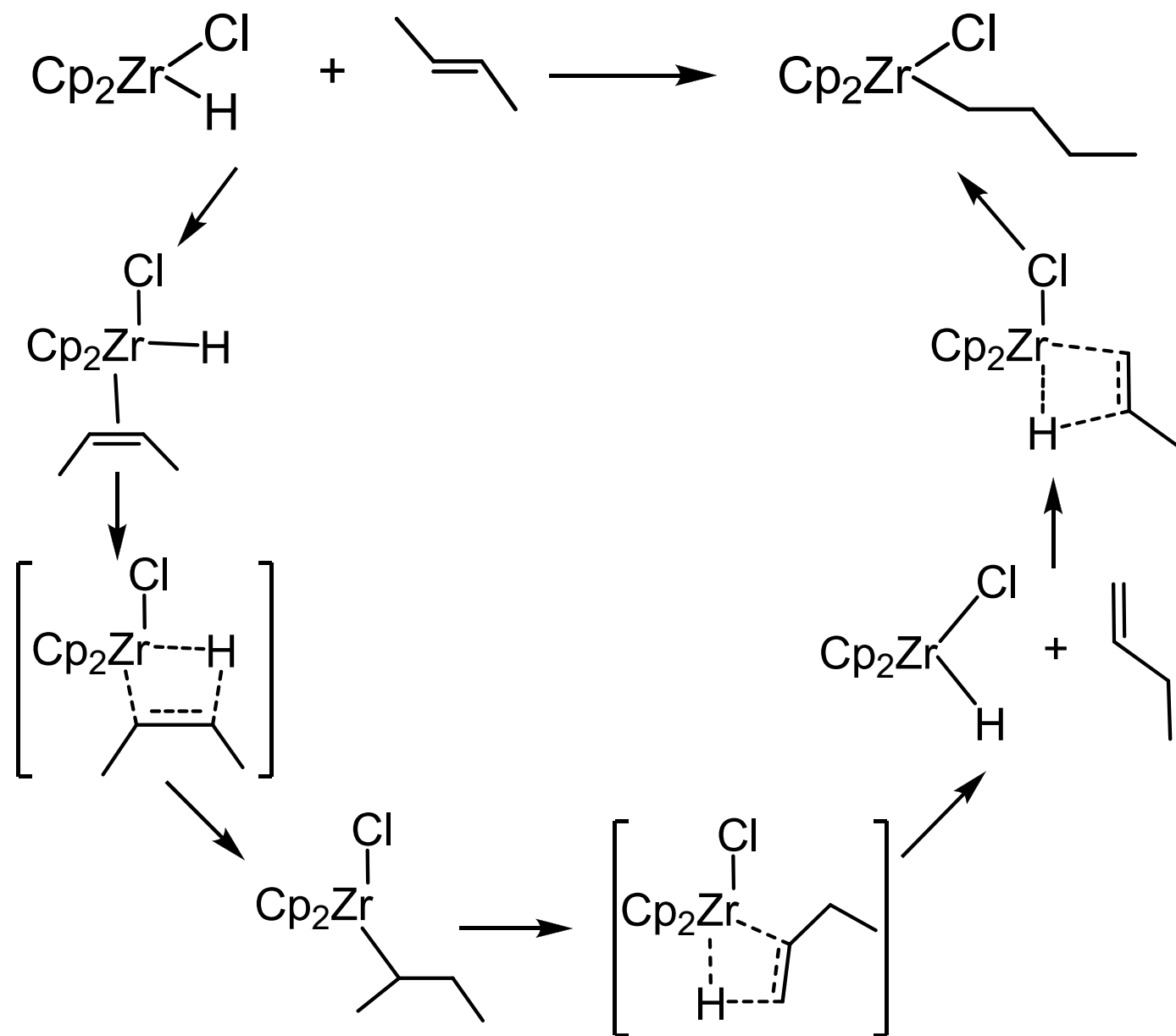
Q1. Propose a mechanism for the following reaction:



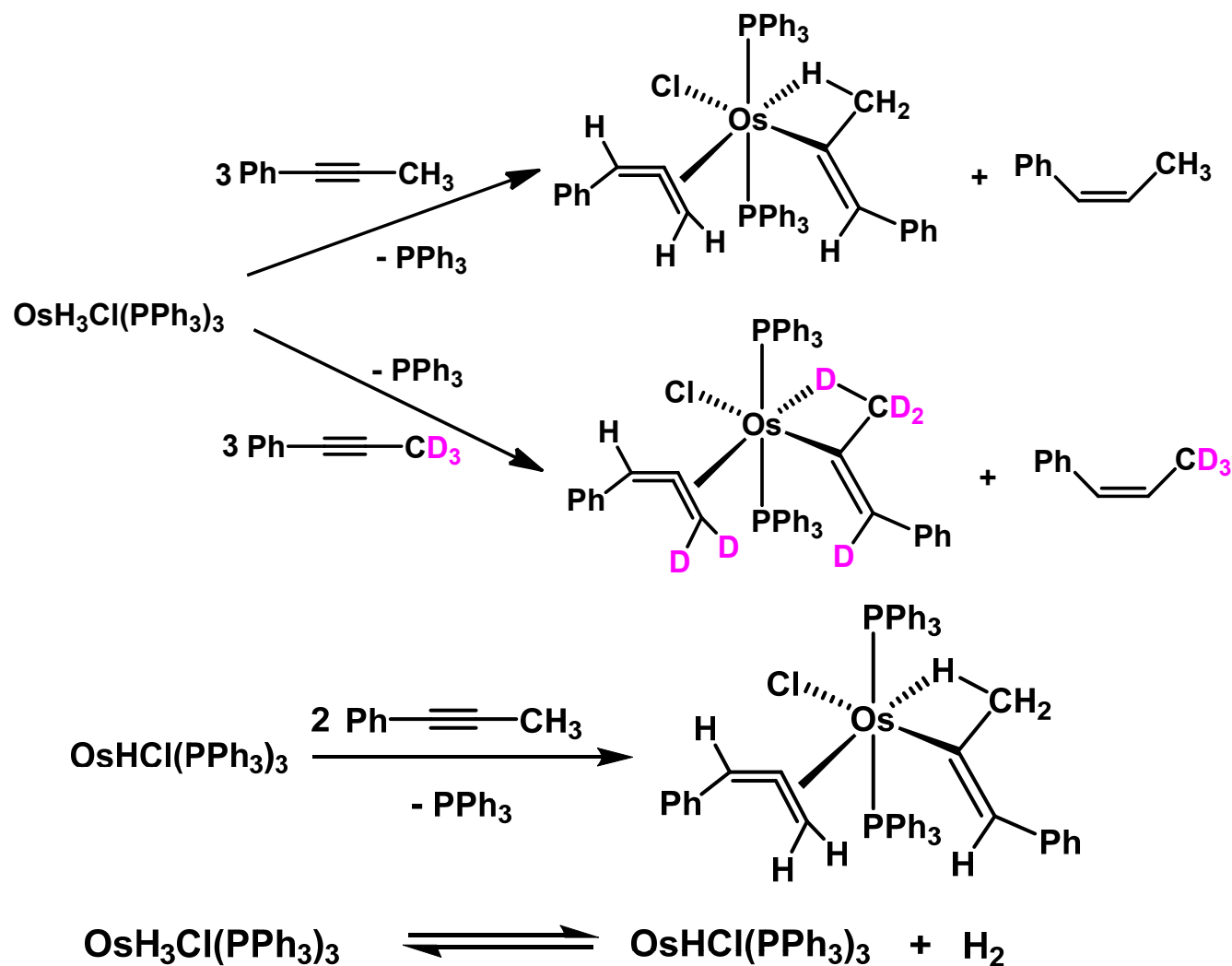
Q1



Q2. Propose a mechanism for the following reaction:

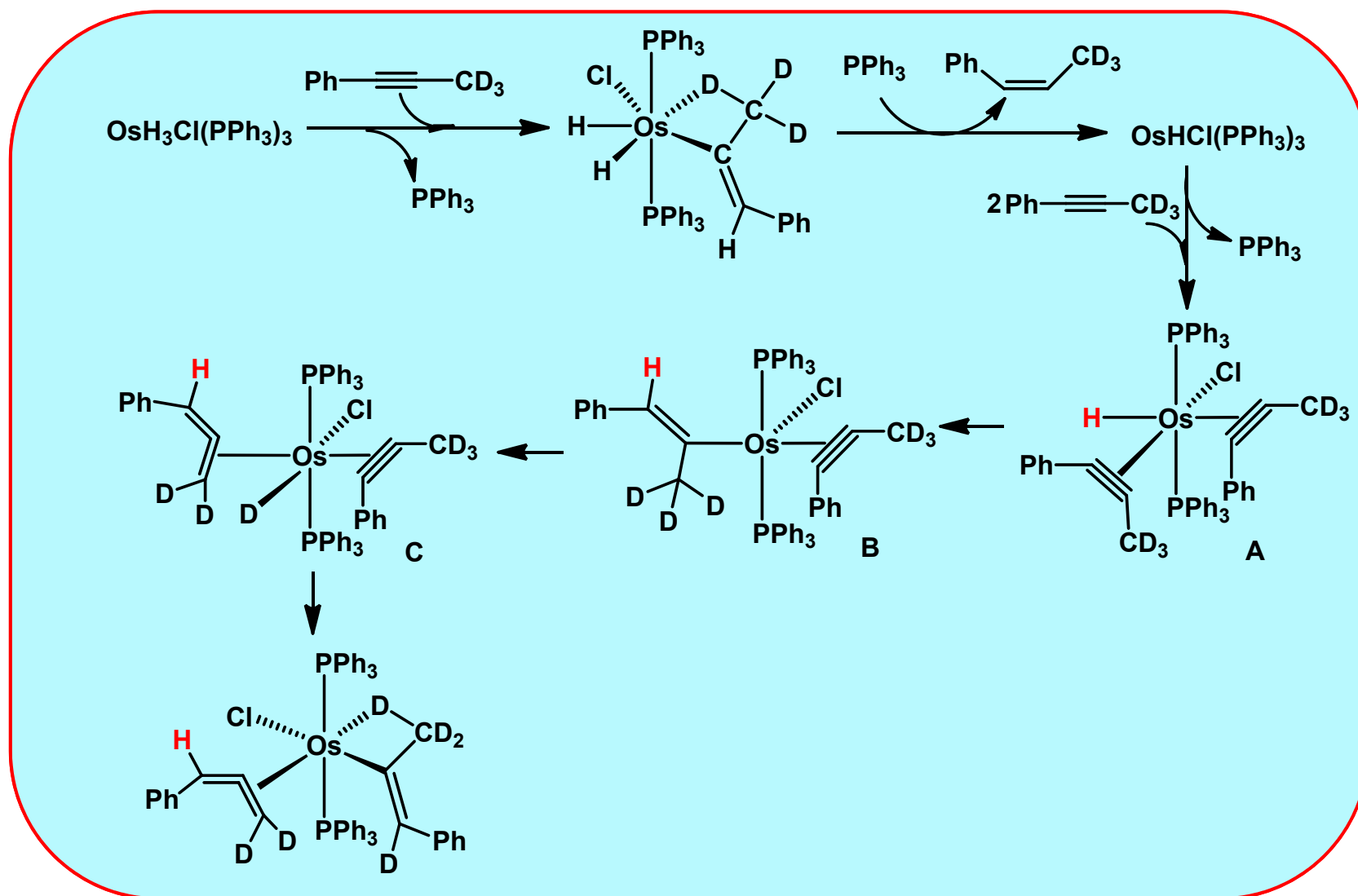


Isomerization of $\text{CH}_3\text{C}\equiv\text{CPh}$ to Phenylallene Promoted by an Osmium Hydride Complex



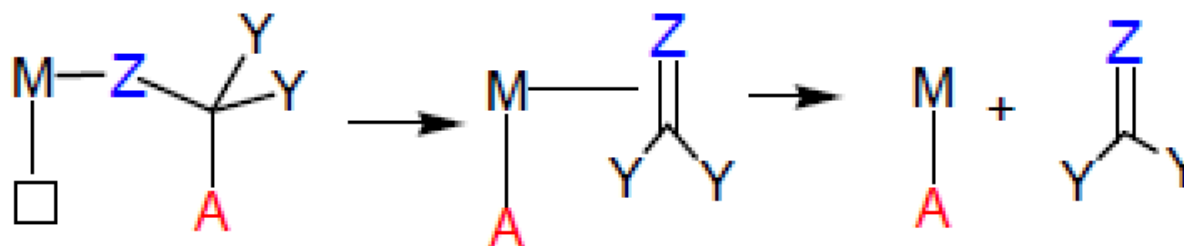
(Wen, T. B., Jia, G. *et al*, *Organometallics*, **2000**, 19, 3466.)

Proposed mechanism for the formation of allene



β-Elimination reactions

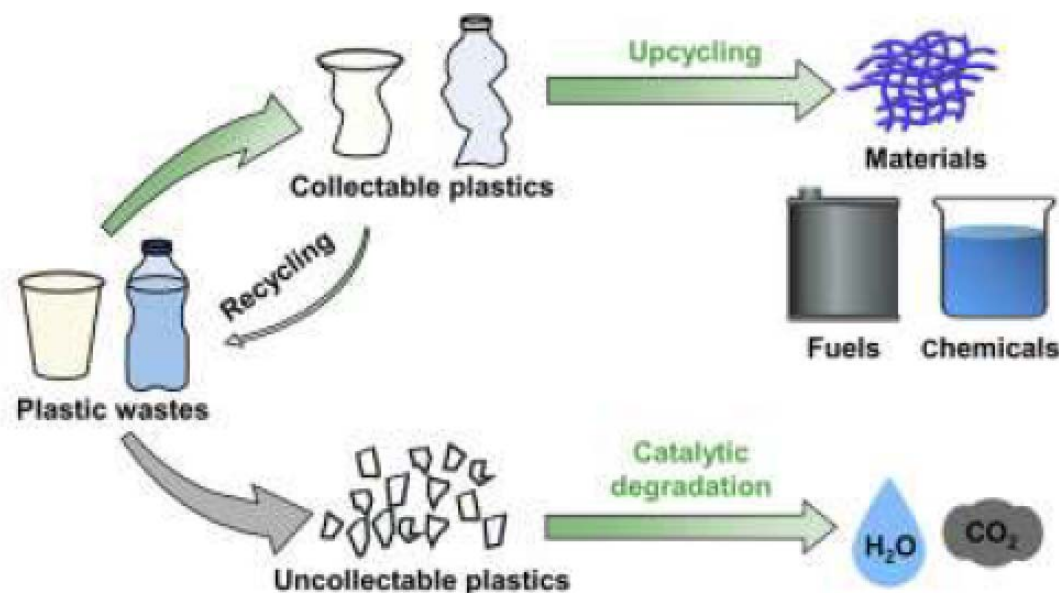
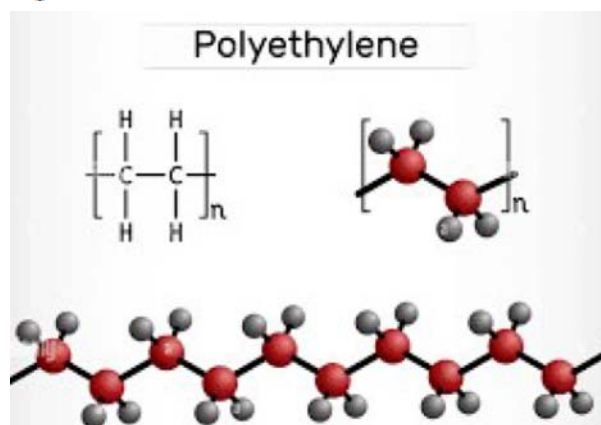
- Other β-elimination reactions are also possible.



e.g. $\text{A} = \text{SiR}_3, \text{BR}_2, \text{halogen}, \text{CR}_3 \dots\dots$

$\text{Z} = \text{CH}_2, \text{CR}_2, \text{O}, \text{S}, \dots\dots$

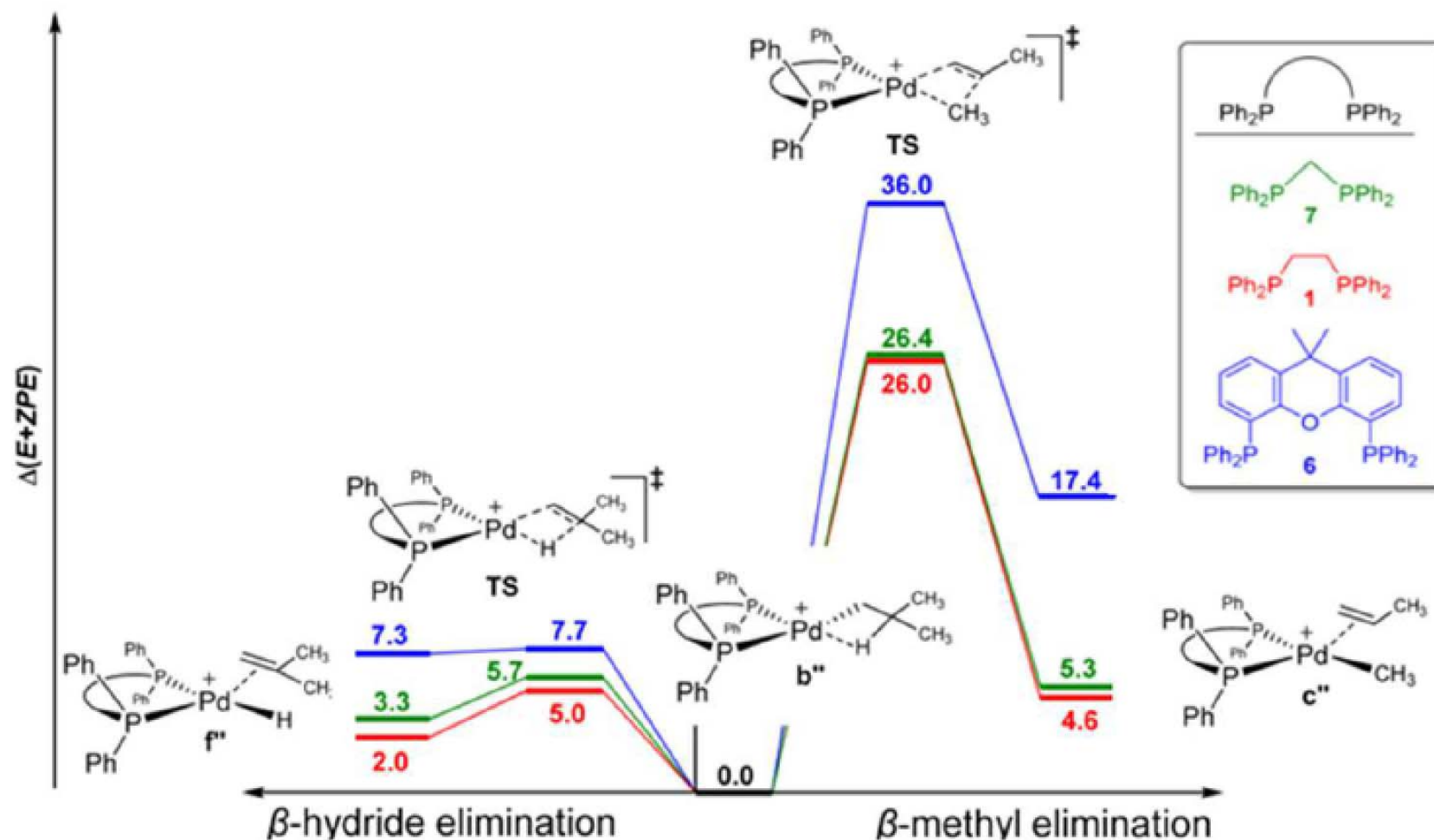
- ✓ **β-Carbon elimination** is of particular interest, e.g. for depolymerization of plastics.



β -Carbon Elimination

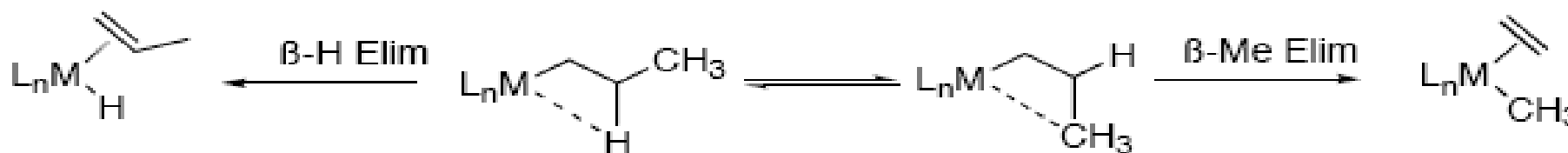
In general, β -alkyl elimination is less favorable than β -H elimination.

Why? An M-H bond is stronger than an M-C bond.



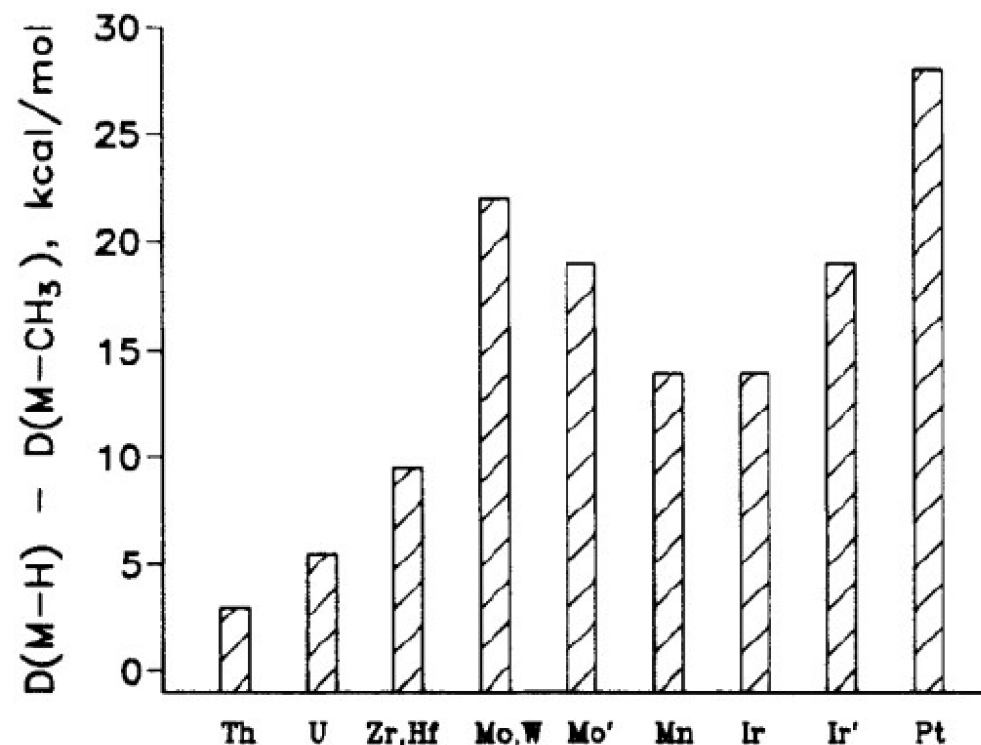
β -Carbon Elimination

In general, β -alkyl elimination is less favorable than β -H elimination.



Why? An M-H bond is stronger than an M-C bond.

Difference in the strength of M-H/M-C

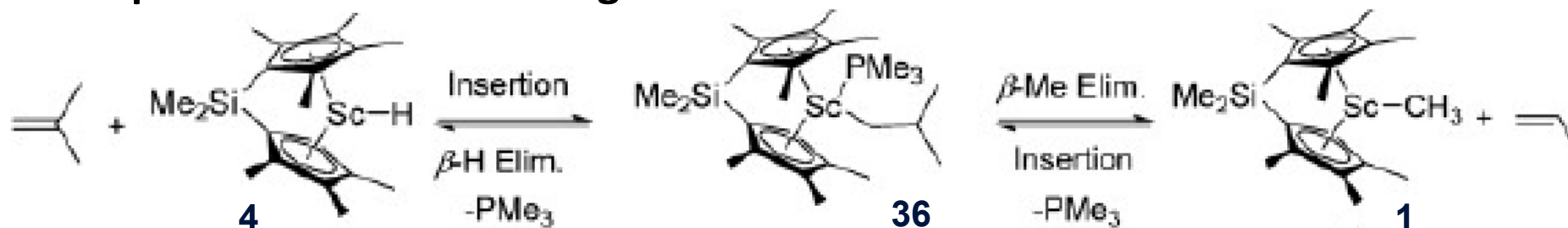


J. Am. Chem. Soc. 1988, 110, 7701-7715.

The difference in the strength of M-H/M-C is less for early and f-block alkyls.

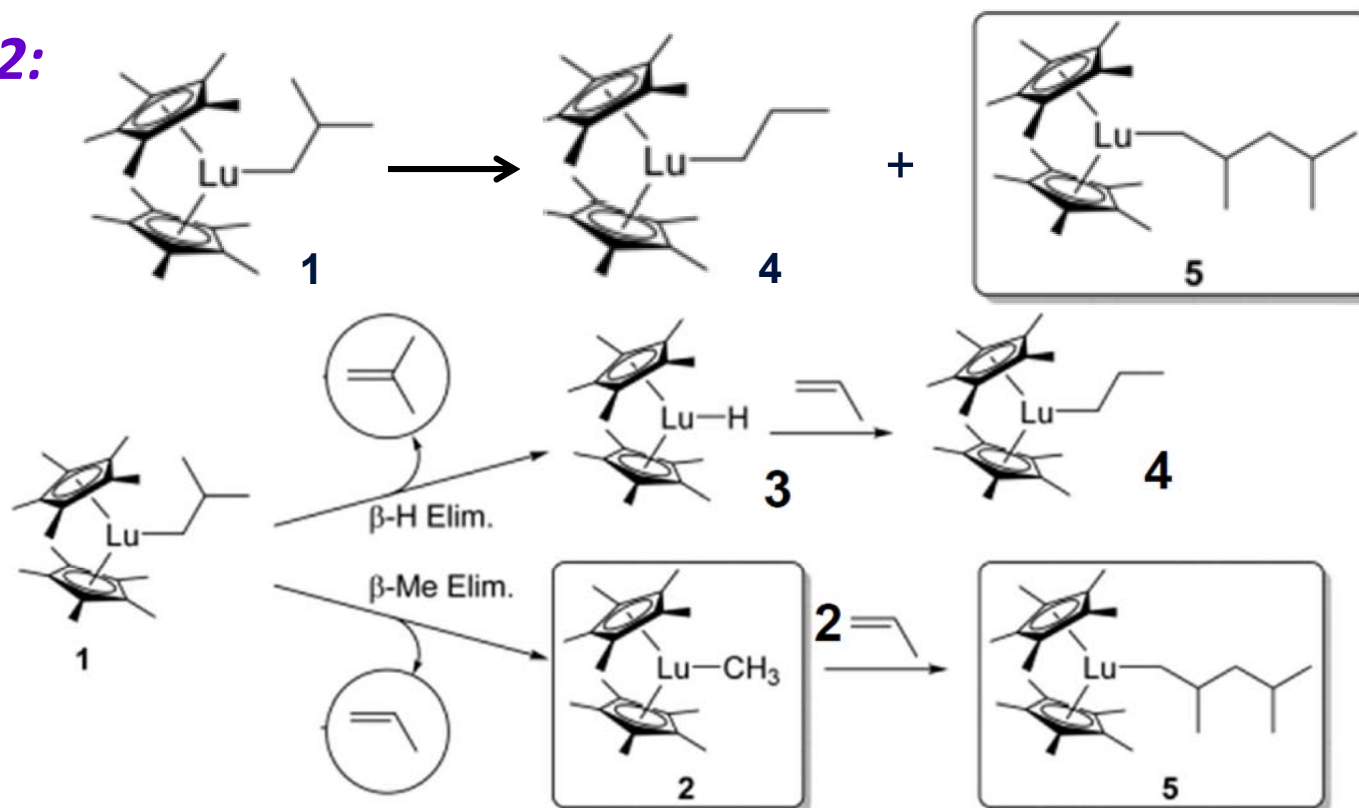
β-Carbon Elimination

- ✓ For early transition metal and f-block alkyls, β-H and β-alkyl elimination may compete with each other. e.g.



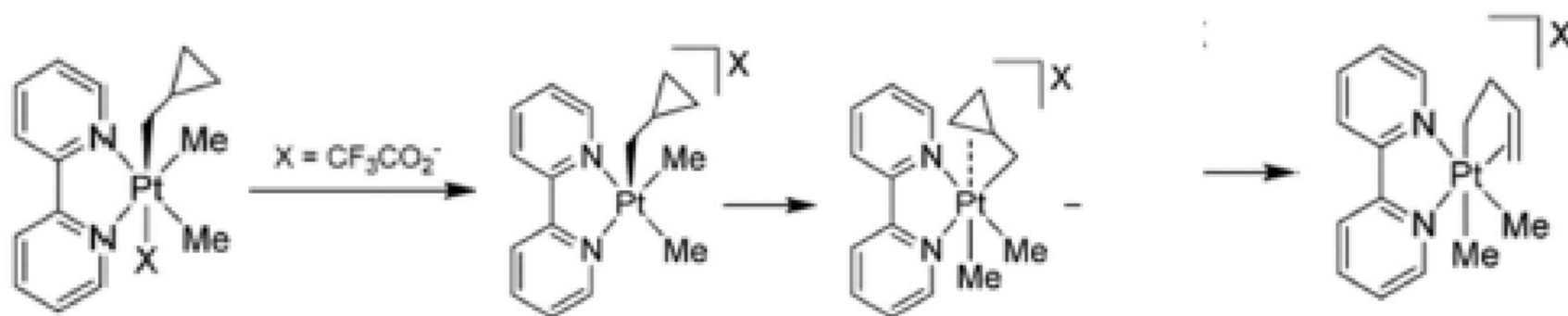
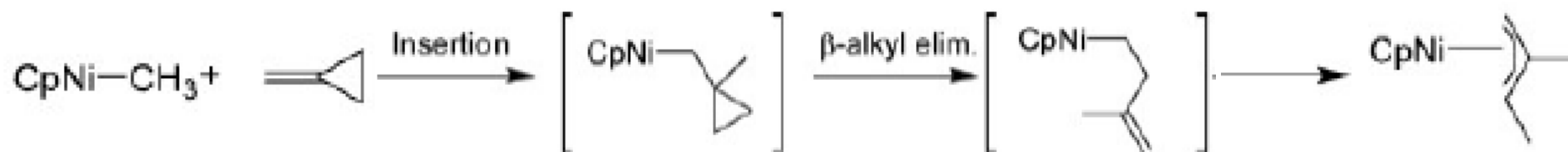
Chem. Rev. **2016**, *116*, 8105–8145

Example 2:



β -Carbon Elimination

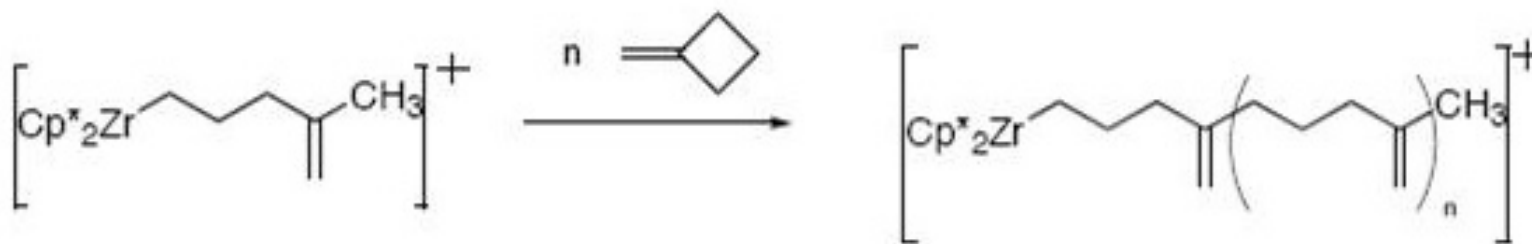
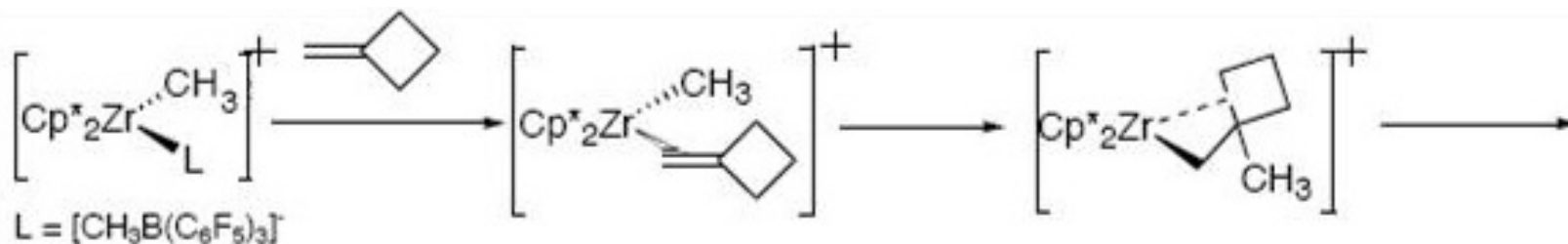
Typically β -alkyl elimination is only observed where there is a large amount of **strain in the C-C bond**, or where there is **no β -H** to undergo elimination. e.g.



Chem. Rev. **2016**, *116*, 8105–8145

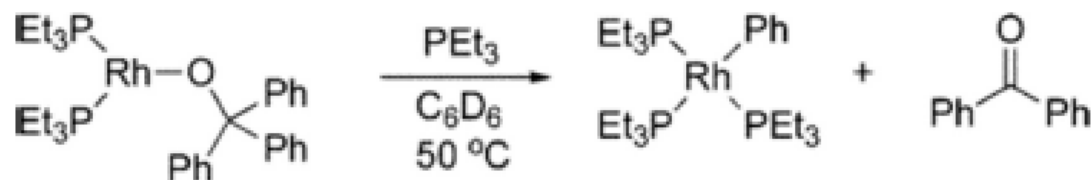
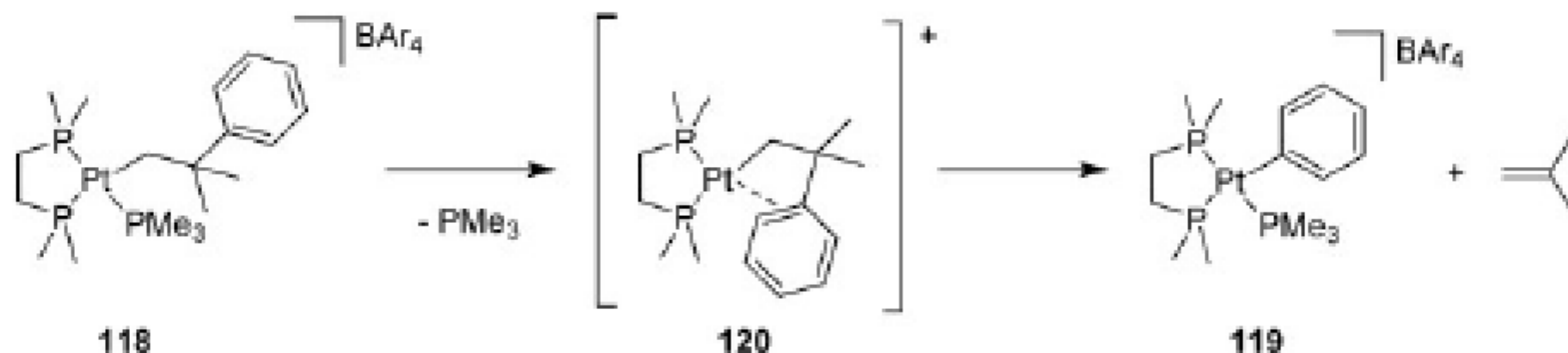
β-Carbon Elimination

β-Alkyl elimination is more common with highly electrophilic d^0 metal complexes.

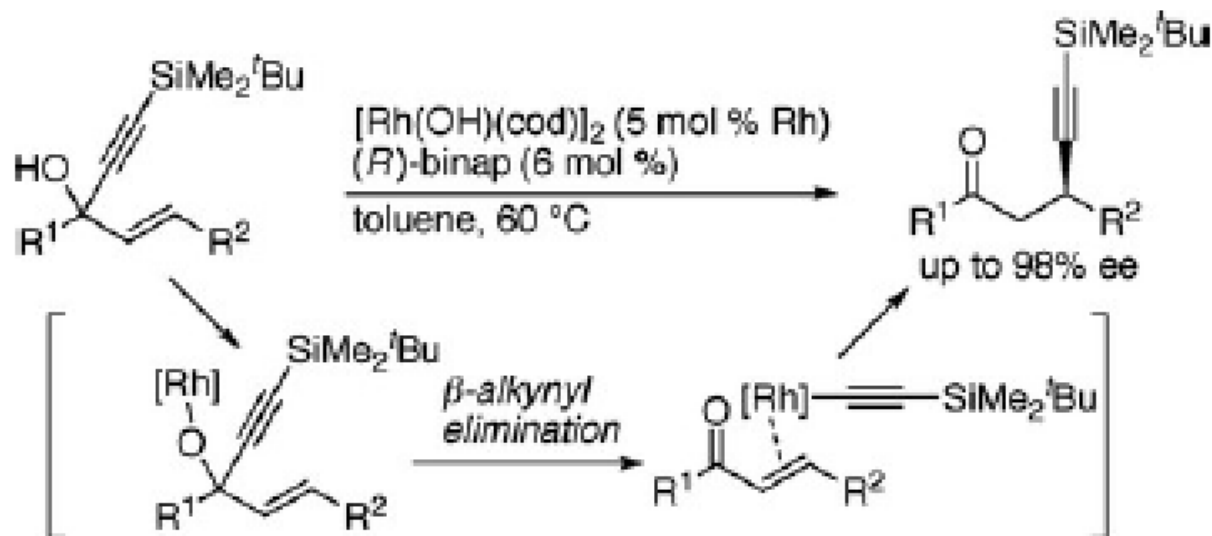


β-Carbon Elimination

- ✓ Additional examples of β-alkyl elimination reactions.



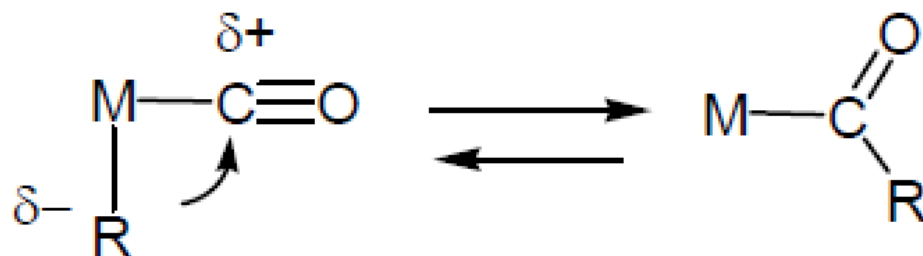
J. Am. Chem. Soc. **2006**,
128, 3124.



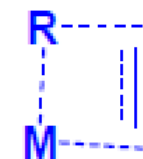
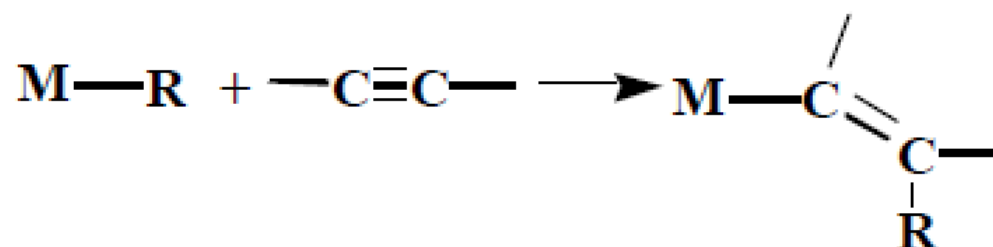
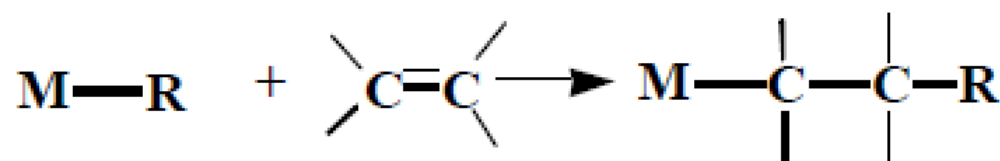
J. Am. Chem. Soc. **2007**,
129, 14158.

Summary

➤ insertion/de-insertion reactions of CO



➤ Insertion reactions of olefins/alkynes and related ligands (e.g. $\text{R}_2\text{C}=\text{O}$, $\text{O}=\text{O}$, $\text{O}=\text{C}=\text{O}$)



➤ Elimination reactions

- α , β , γ and δ -elimination reactions.
- β -H elimination reactions are very common.